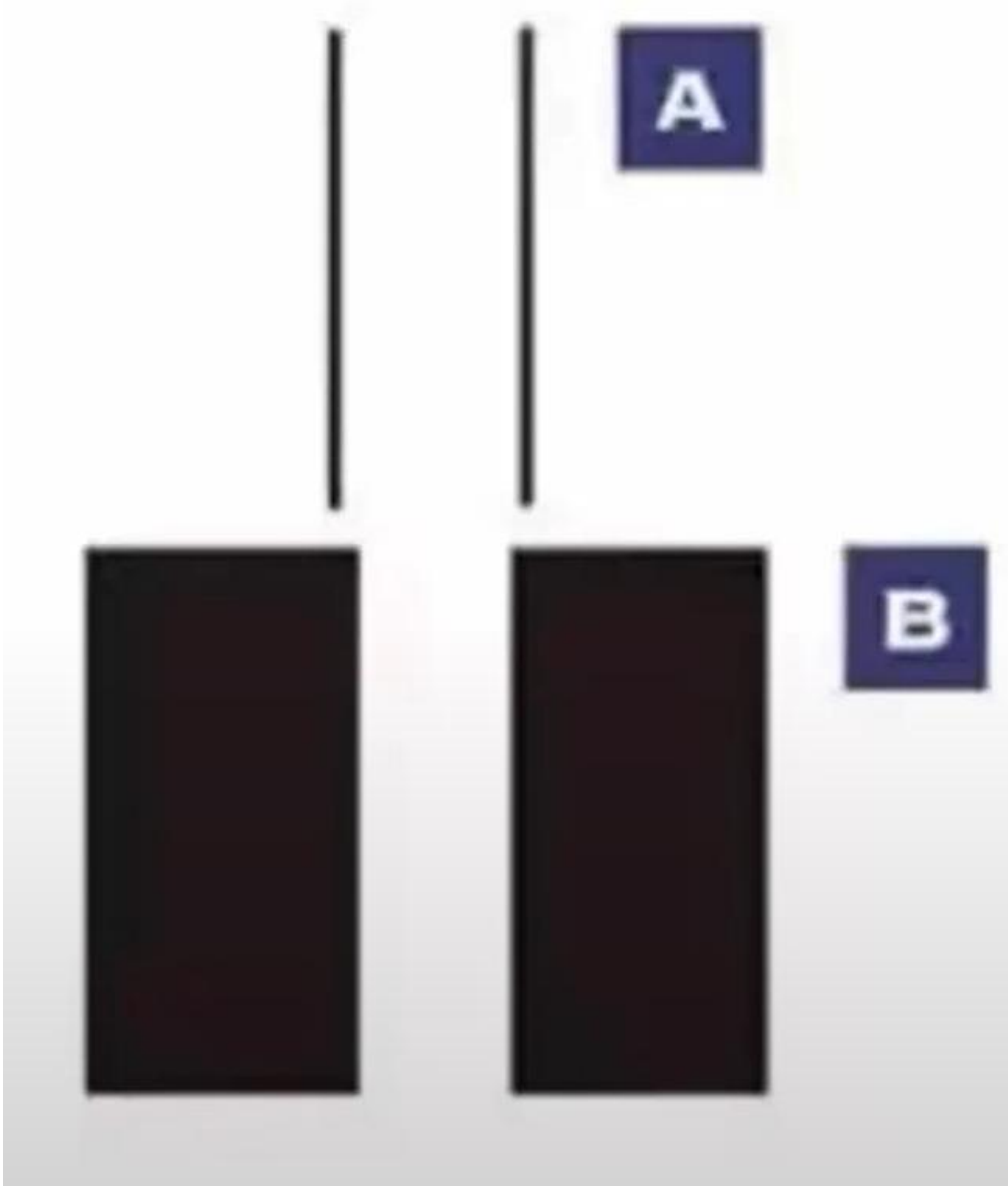
計算認知神經科學
腦與感官



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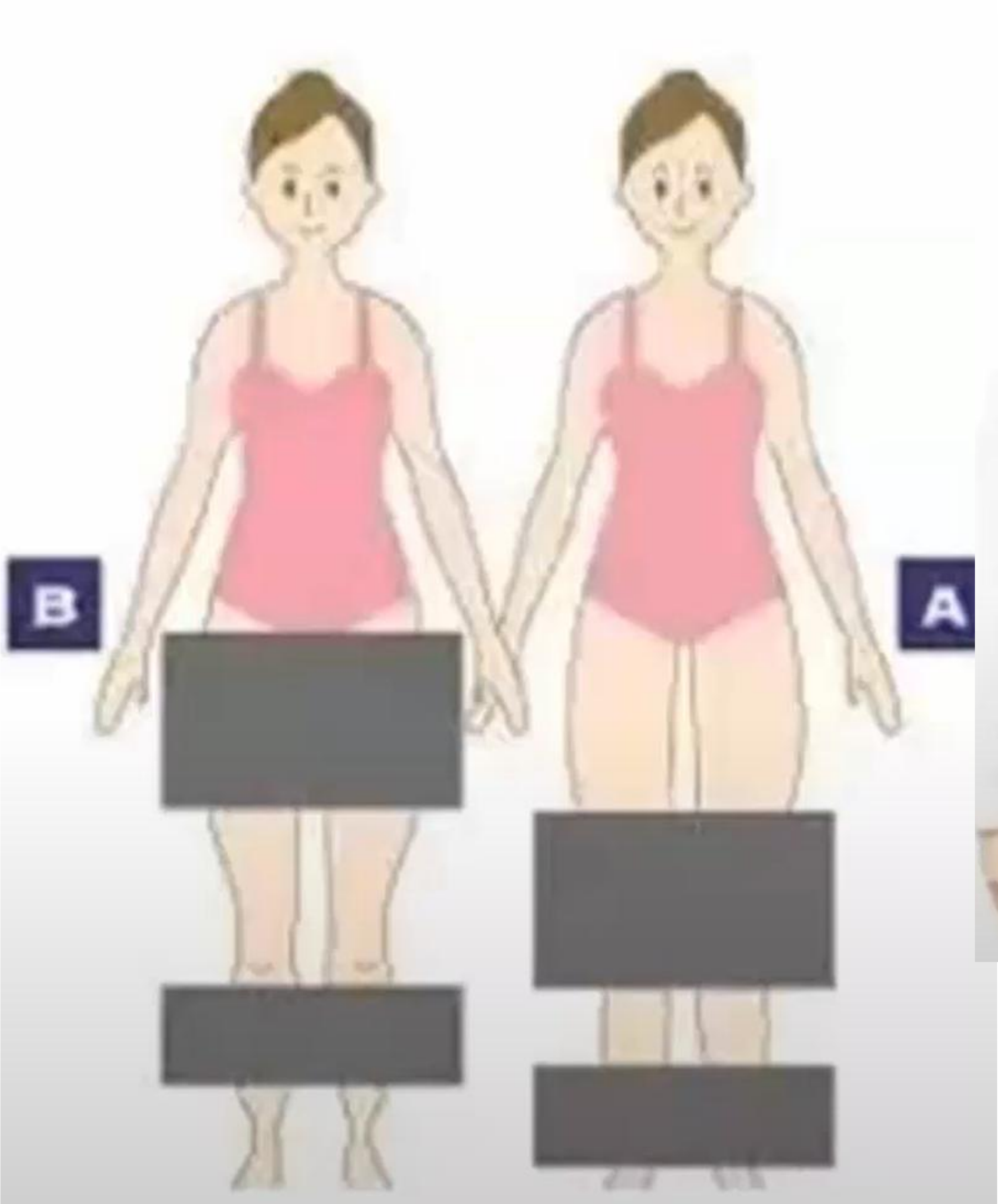

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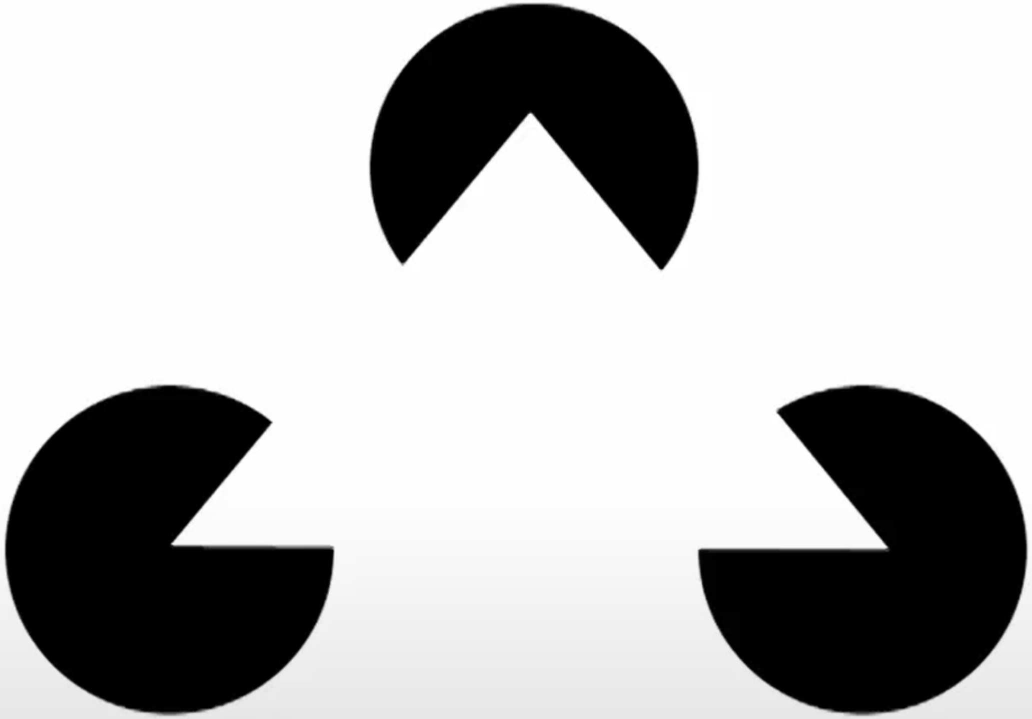
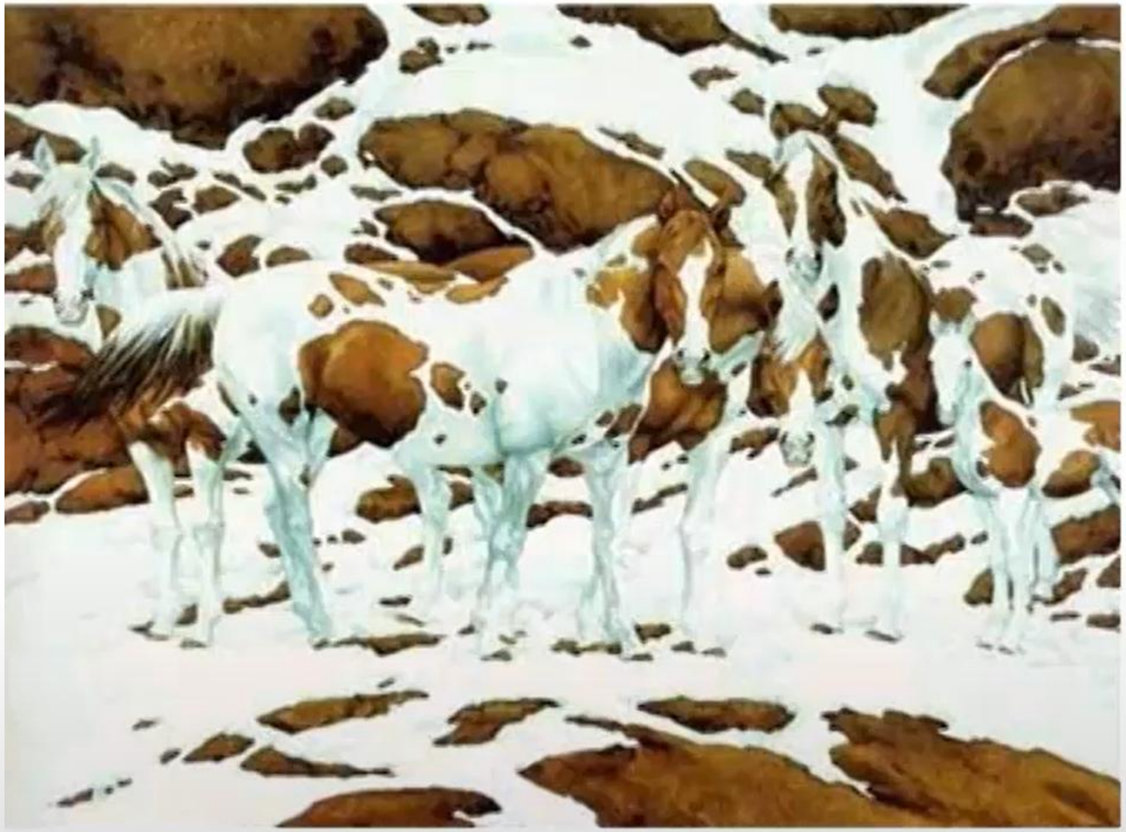
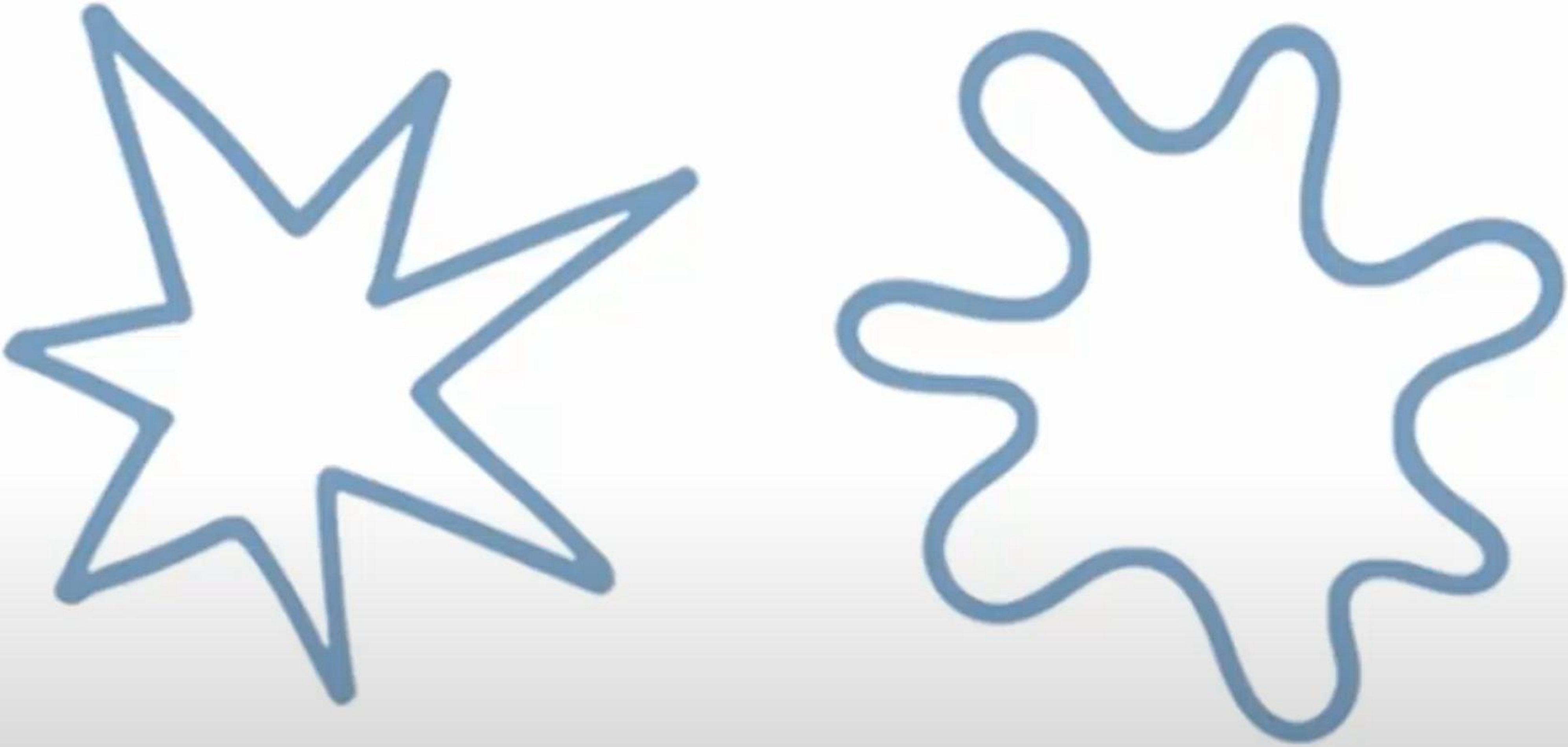


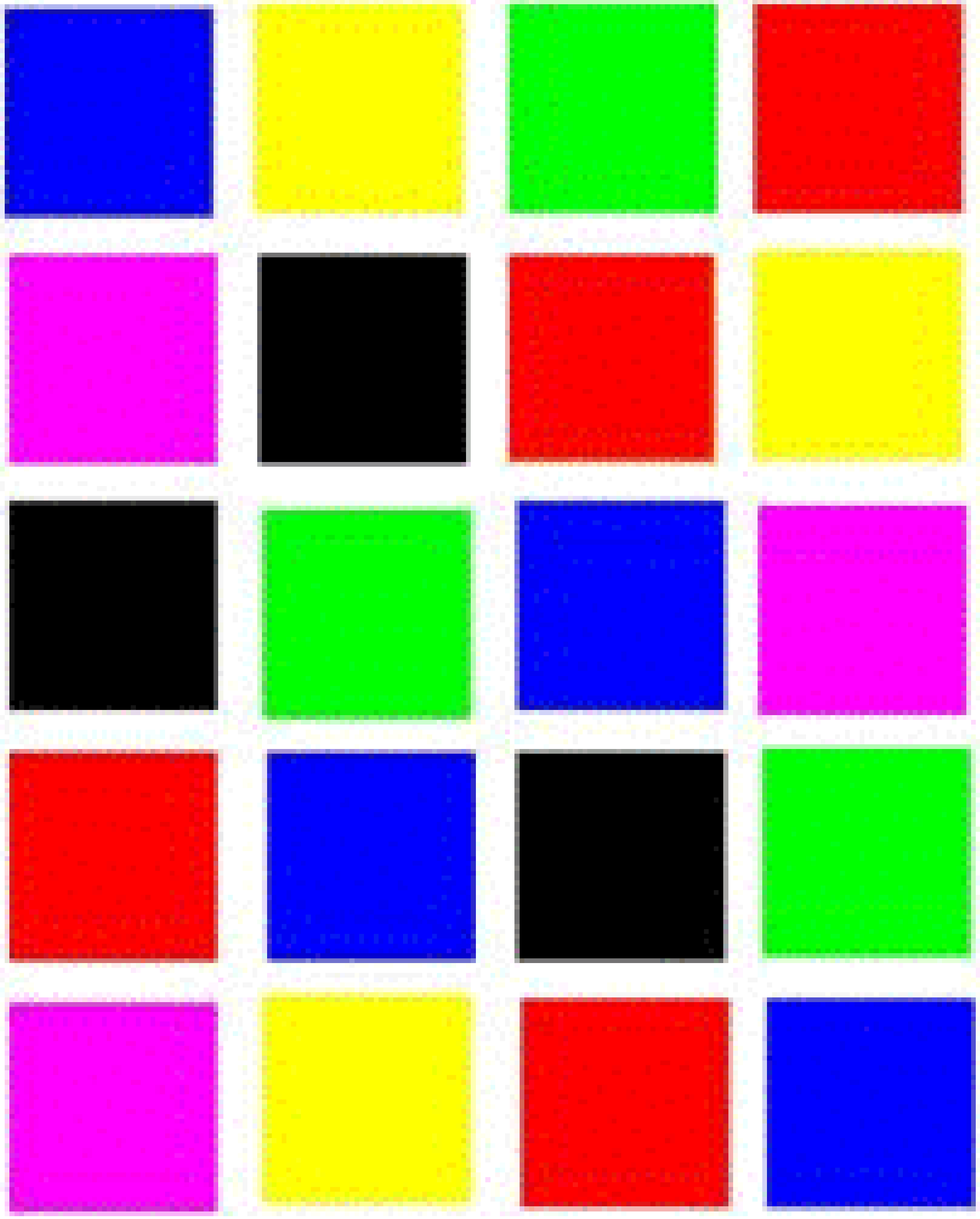


Are face special?

Who is Kiki? Who is Bobua?



史楚普叫色實驗



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史楚普叫色實驗

Psychological experiment: Color-word Stroop task

- Slower response for incongruent trials (the color of a word is not the meaning of the word).
- Semantic process is automatic.
- You need to actively suppress the information processing of semantic knowledge.

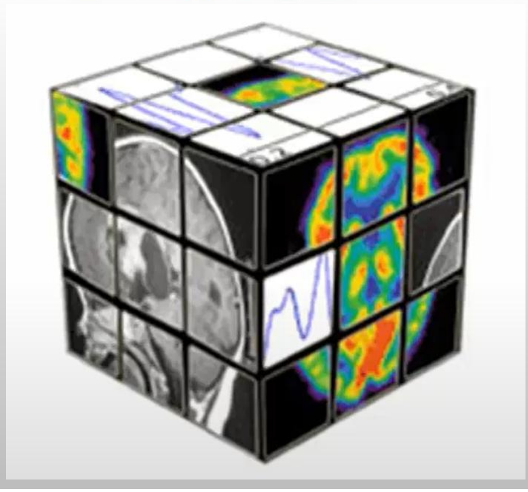
紅	紅
Congruent trial	Incongruent trial

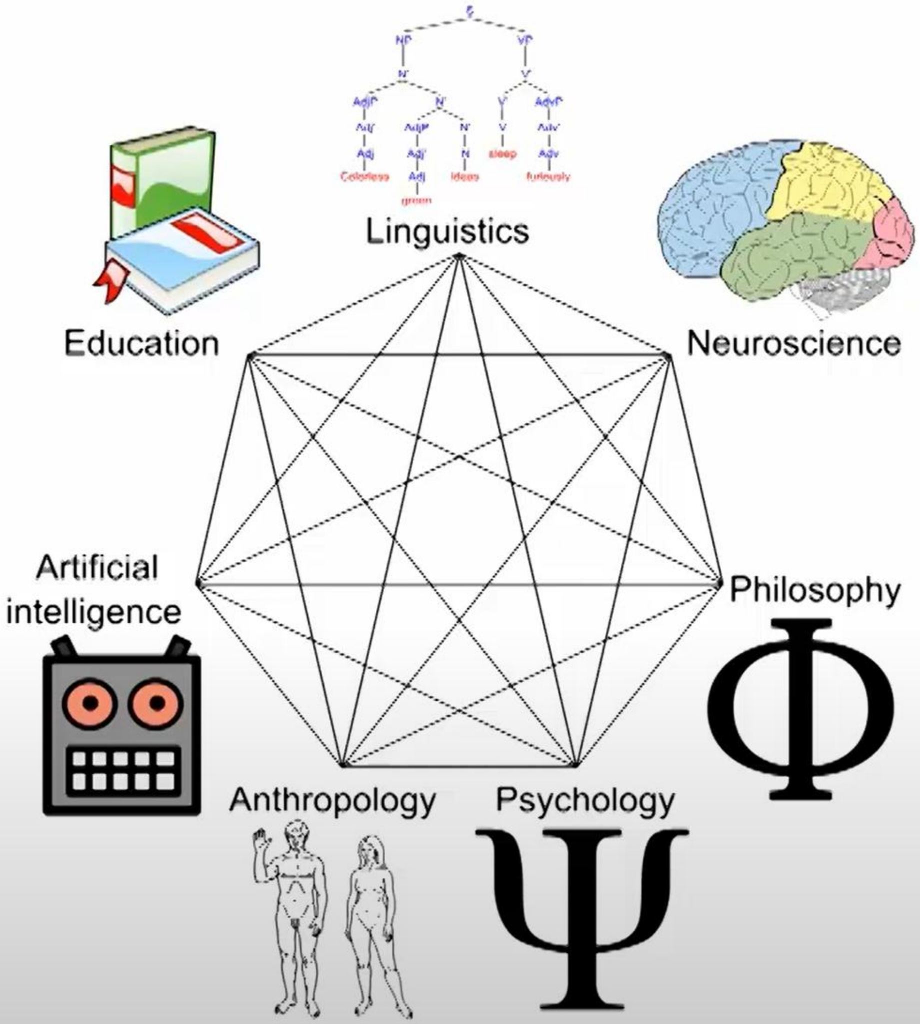
Topics in cognitive neuroscience
(incomplete list)

- Sensation & Perception
- Object Recognition
- Control of Action
- Learning and Memory
- Emotion
- Attention & Consciousness
- Language comprehension & Production
- Thinking & Decision Making
- Plasticity
- Social cognition
- And so on...

What is cognitive science?

- Interdisciplinary scientific study of the mind and its processes
- Examines:
  - ➢ what cognition is
  - ➢ what cognition does
  - ➢ how cognition works





https://apps.carleton.edu/curricular/cgsc/

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What is cognitive neuroscience (Brain and Cognitive Science)?

Cognitive Neuroscience is the marriage of neuroscience (studying the underlying brain mechanisms) and cognitive psychology (studying cognition at the behavioral level).

- Neuroscience:
  - ✓ Varieties of Species
  - ✓ Level of Analysis
- Cognitive Psychology
  - ✓ Studying cognition at the behavioral level
  - ✓ mental processes: thinking, perceiving, imaging, speaking, acting, and planning

Representations (表徵)

- Definition: physical properties of the world (e.g., sounds, colors) and abstract forms of knowledge (e.g. knowledge of the beliefs of other people, factual knowledge).

- Representations are manifested in:
  - Cognitive system: mental representation (cognitive psychology)
  - Neural system: neural representation (neuroscience)

Representations (表徵)

- Cognitive system:
  mental representation
  - ➢ Mental model of the outside world
    - ✓e.g., colors, objects, names

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Universal design

the applications of mental representation

http://a-chien.blogspot.tw/2012/11/blog-post_4.html

Universal design

the applications of mental representation

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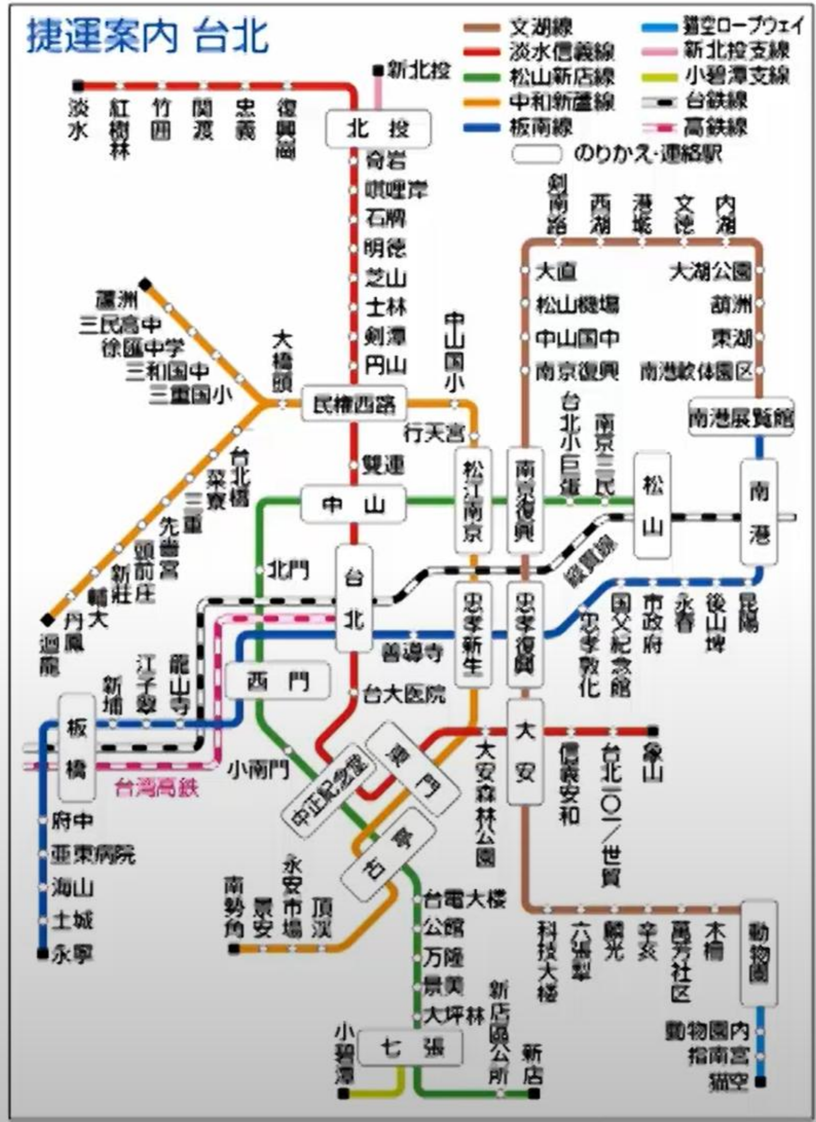
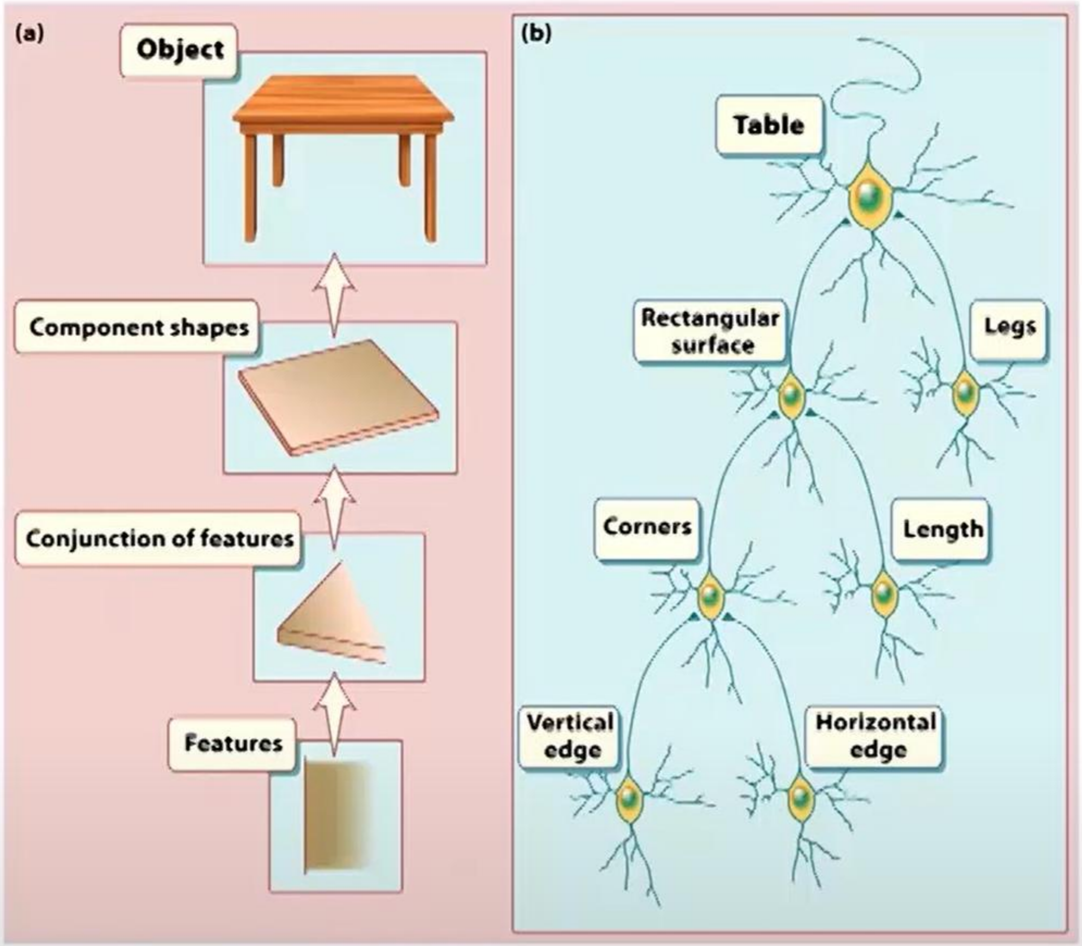
Representations is NOT identical

Taipei metro map in Taiwan

Taipei metro map in Japan



Representations



- Neural system

  Neural representation

  - ➢ The properties of the outside world manifest themselves in the neural signal / biological system
  - ➢ e.g., edges, rectangle

The basic concept of images

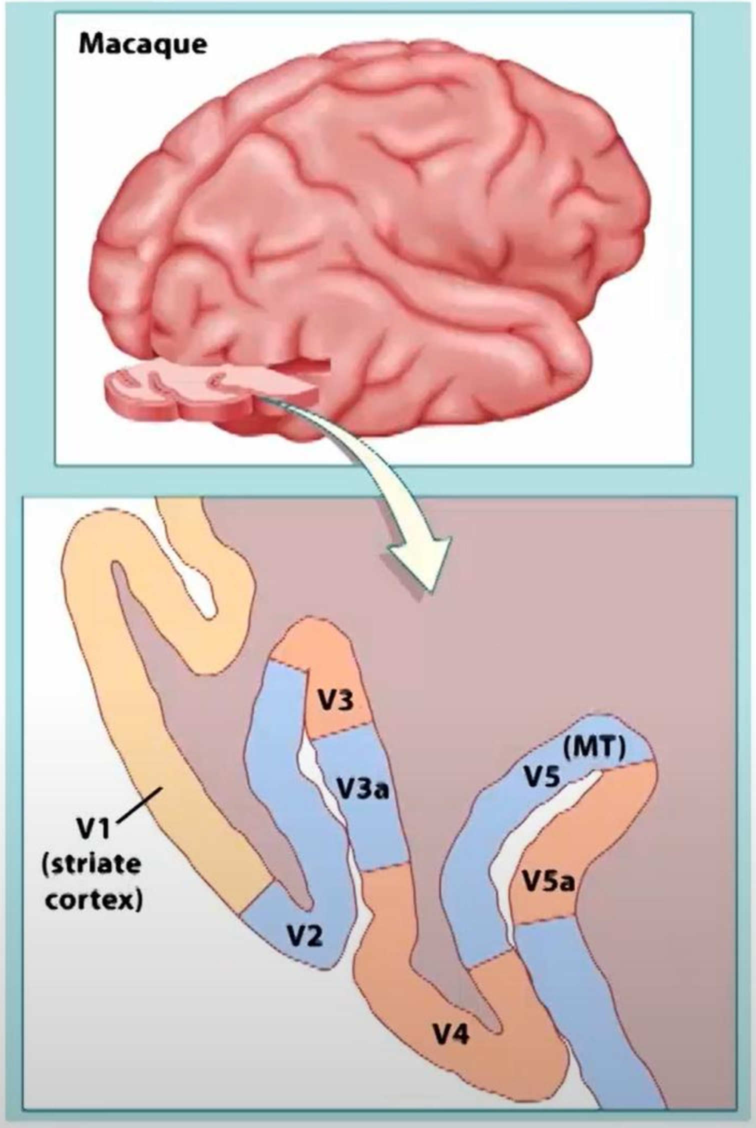
- Before any image processing, we need to understand the “format” of the image.
- Neural images are “digital” images.

low resolution

high resolution

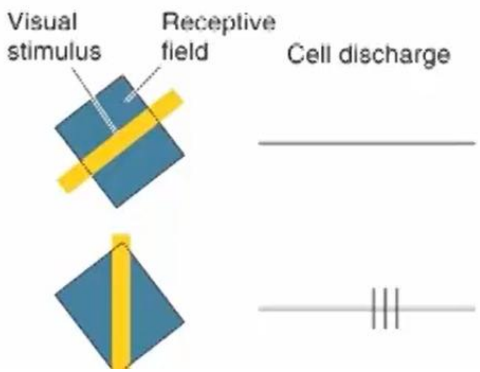
Occipital lobe: cortical visual areas

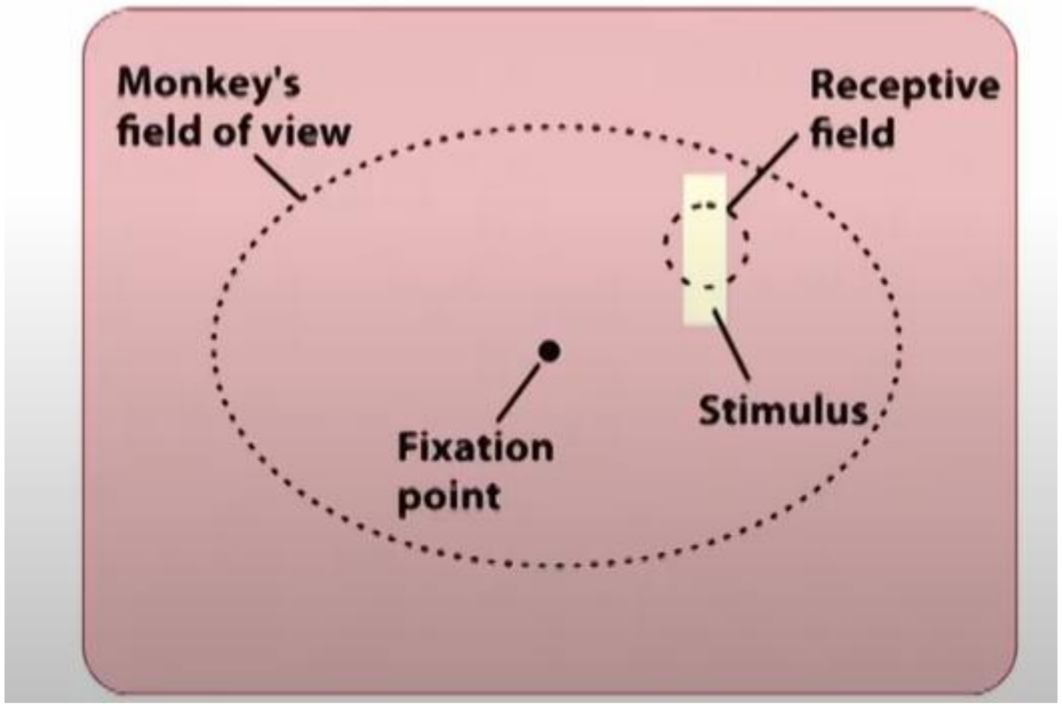
- Map of cortical areas:
  - ➢ Microelectrodes inserted in cortical areas show many areas responding to visual stimuli (in monkeys)
    - ✓ Labeled V1, V2, V3, V4, etc.
  - ➢ Neurons in different areas are sensitive to different types of stimuli



Occipital lobe: cortical visual areas

- A functional specialization of occipital visual areas
  - V1
    - ✓ Optimal stimuli: bars, edges, directions, etc.
    - ✓ Partial separation between eyes
    - ✓ Receptive fields: small



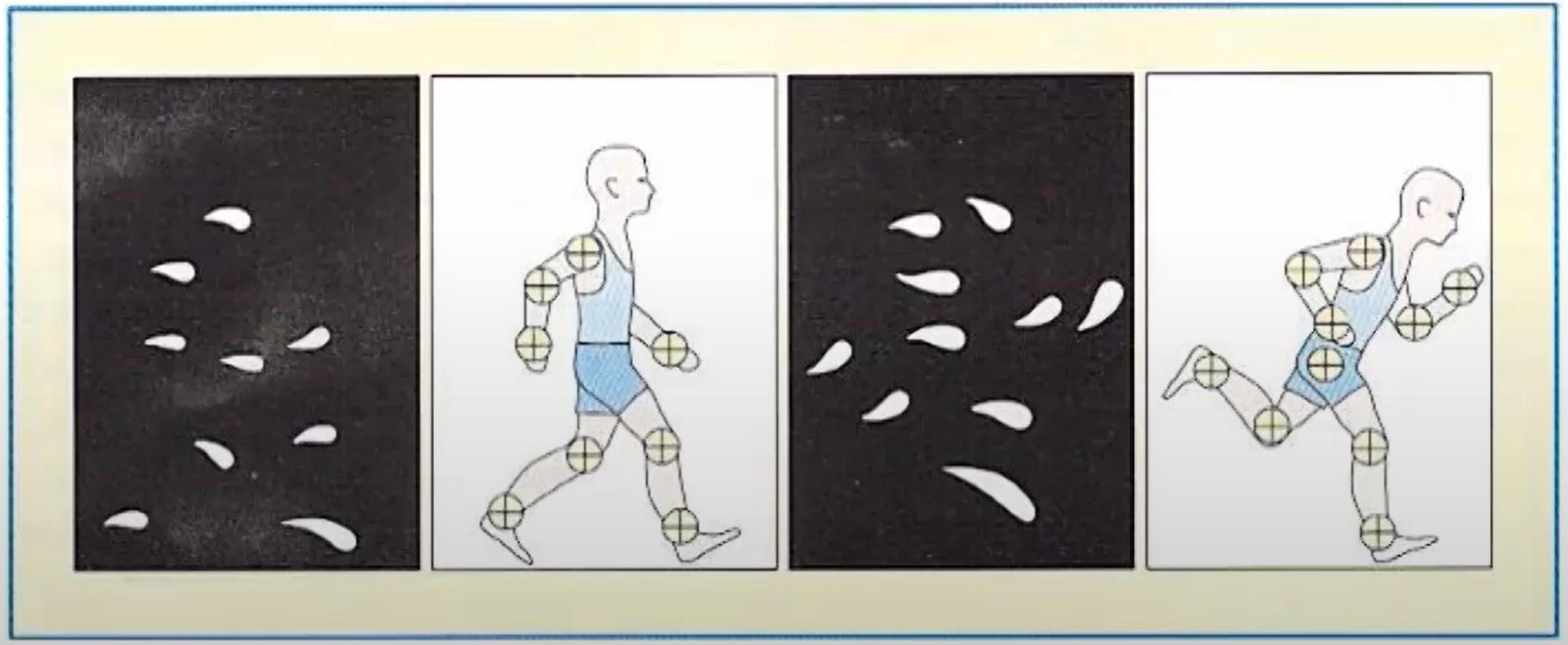


Occipital lobe: cortical visual areas

- A functional specialization of occipital visual areas
  - V2-V4 (visual areas 2-4)
    - ✓ Optimal stimuli: Same as V1, plus moving bars
    - ✓ Most neurons receive input from both eyes
    - ✓ Receptive fields: progressively larger
  - V4: color center of the brain

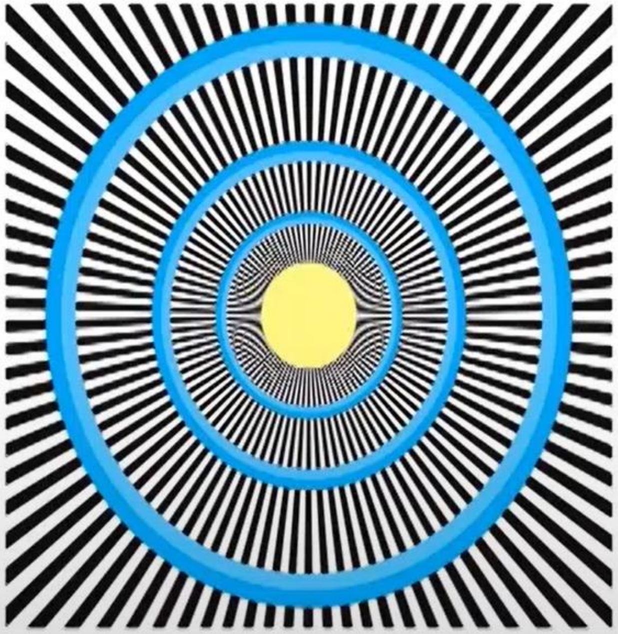
Occipital lobe: cortical visual areas

- A functional specialization of occipital visual areas
  - V5 (also called MT, border with temporal lobe)
    - ✓ Movement center of the brain
    - ✓ Optimal stimuli: Only sensitive to moving object
    - ✓ Receptive fields: very big

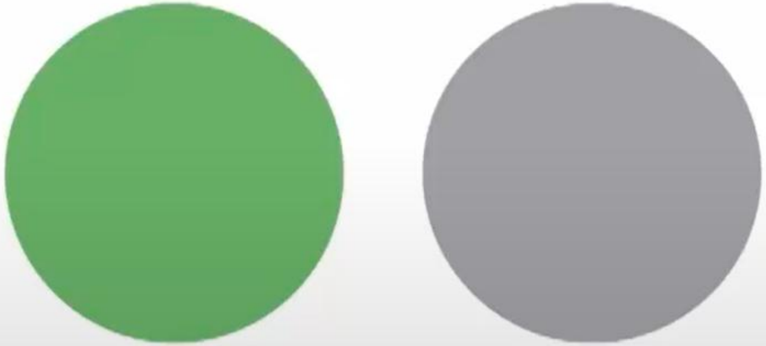


Visual illusions

- Our percepts are more closely related to neural activity in higher visual areas.

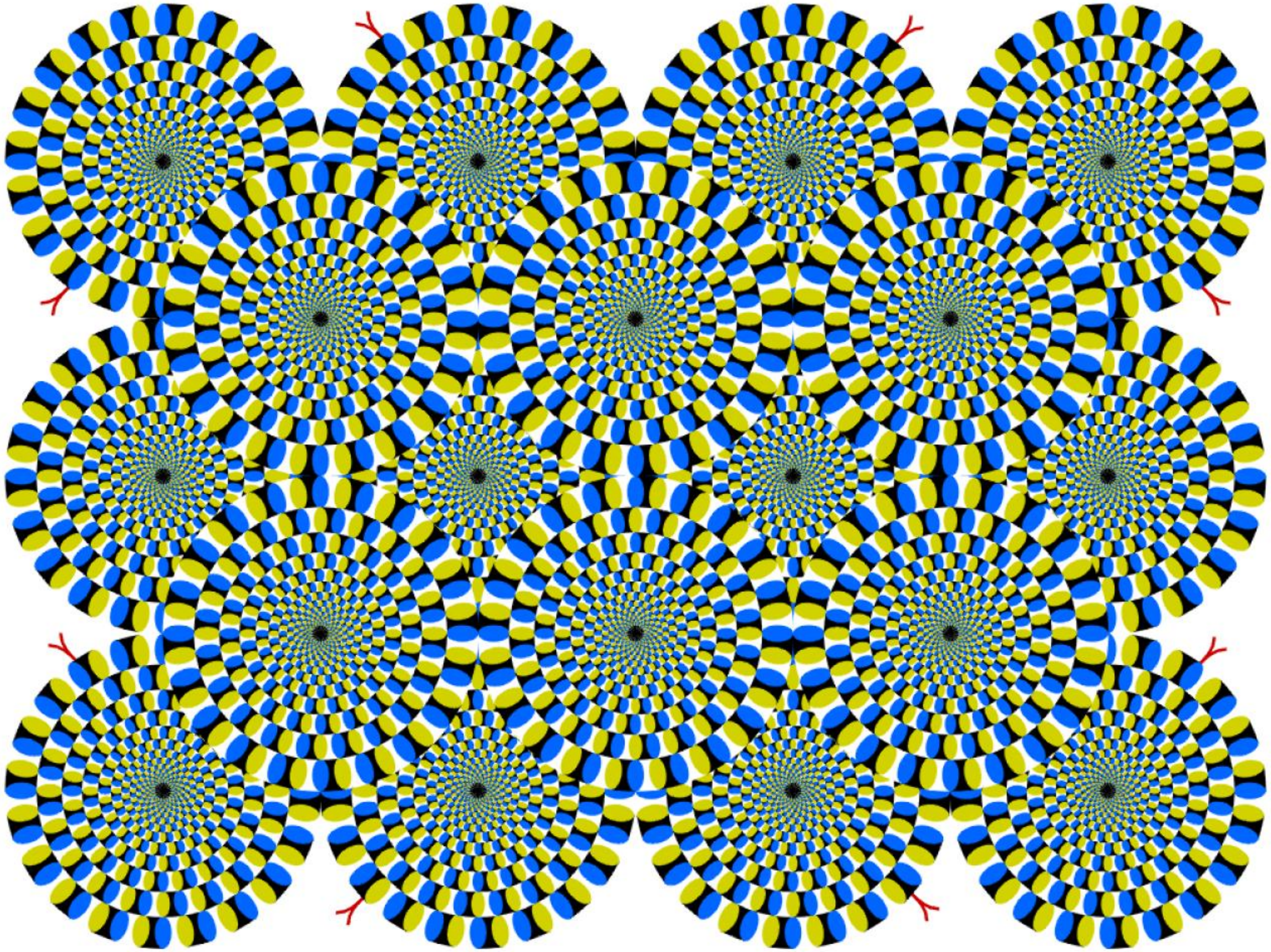


Illusory motion:
activation in area MT



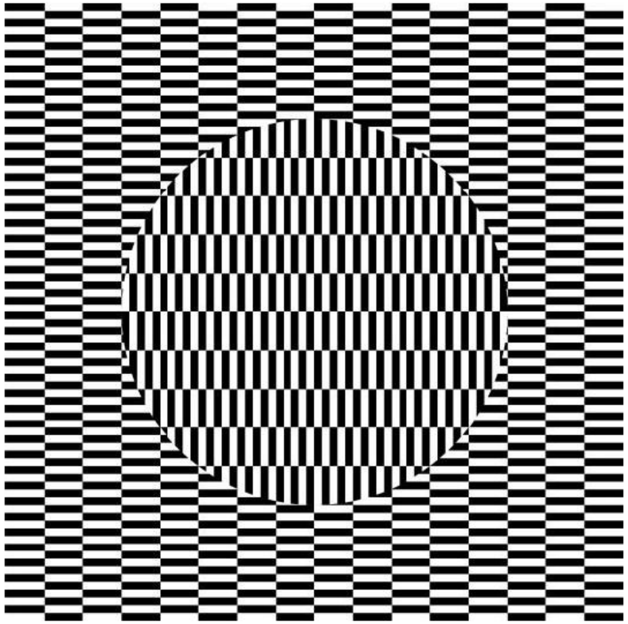
Color aftereffects:
activation in area V4

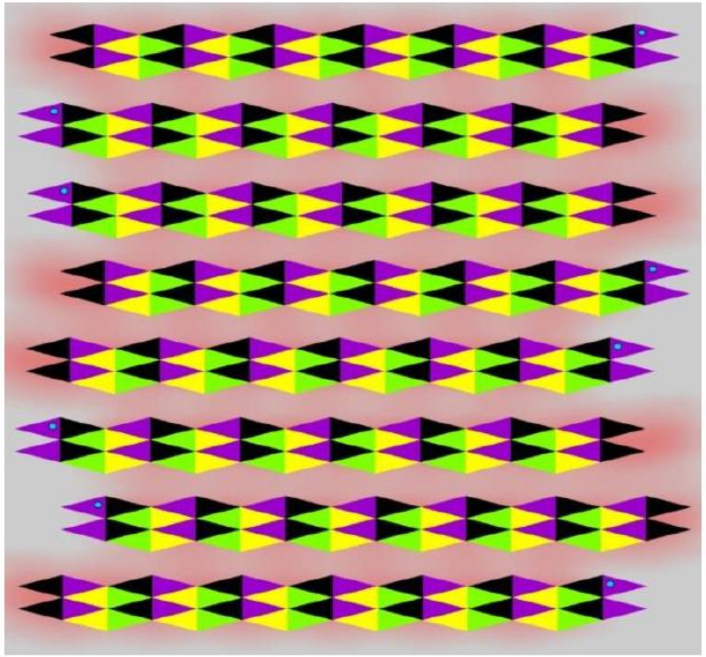
Seeing motion from a static picture?

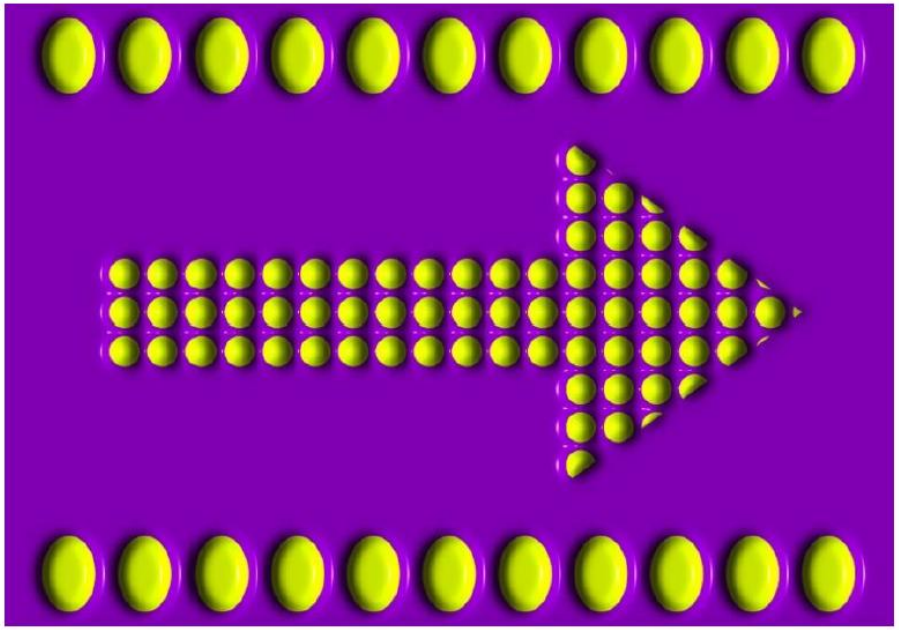


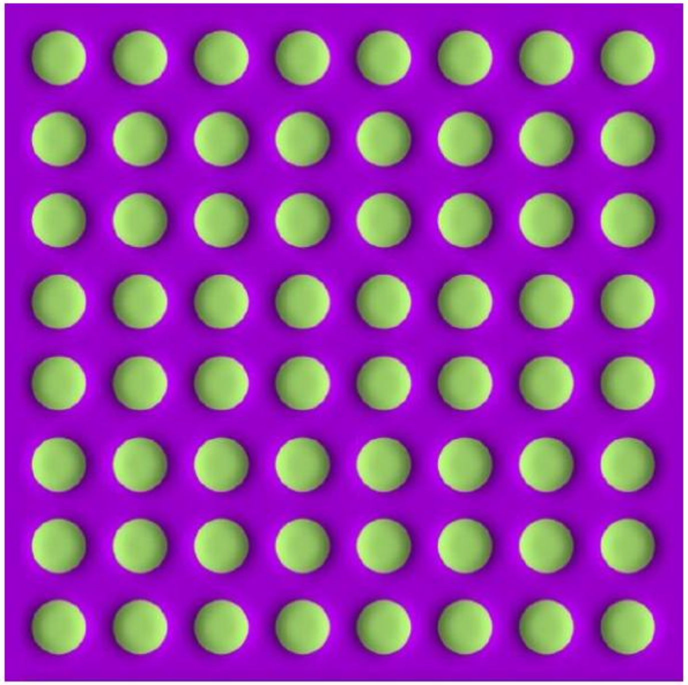
http://www.ritsumei.ac.jp/~akitaoka/index-e.html

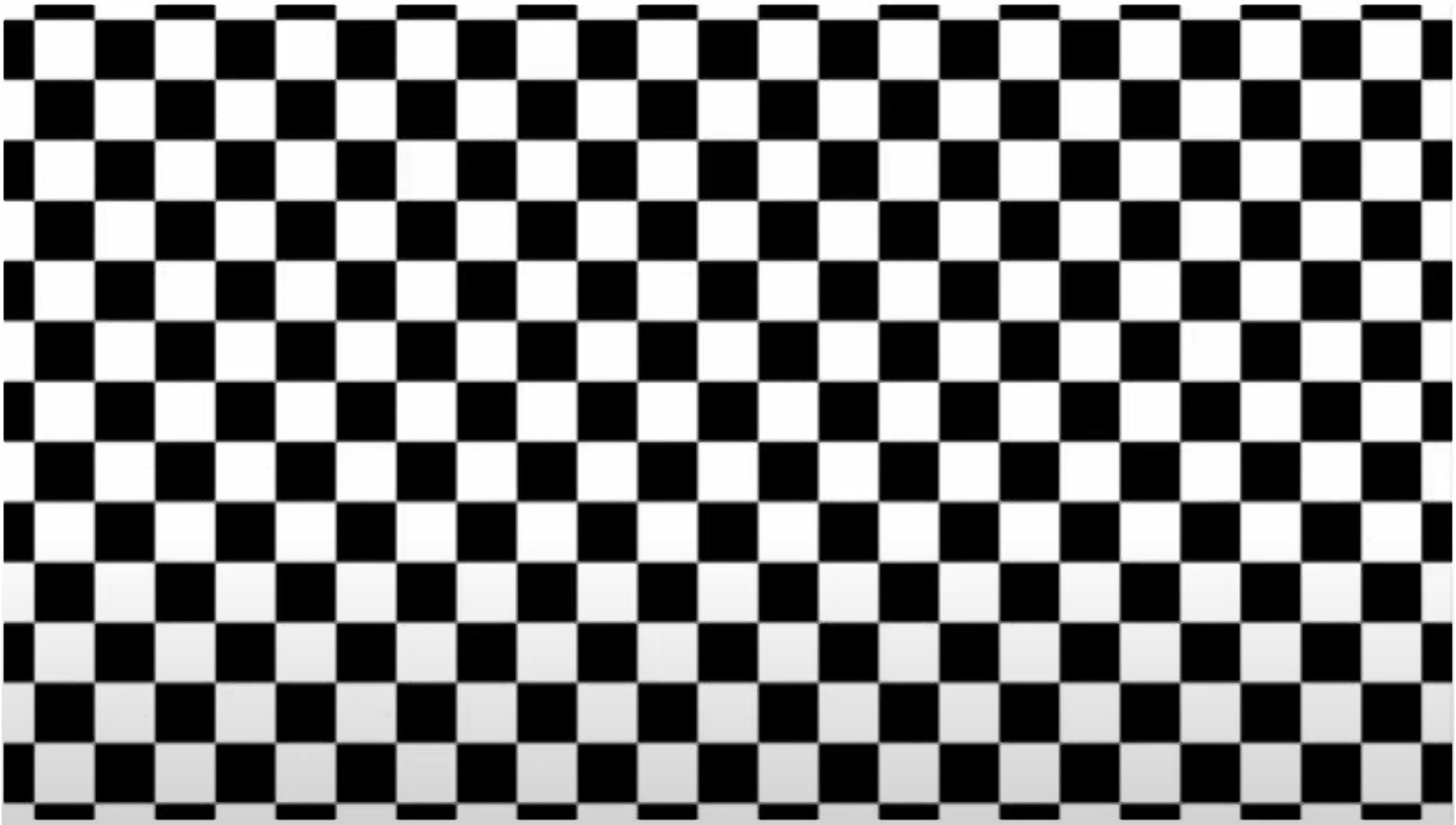
More examples











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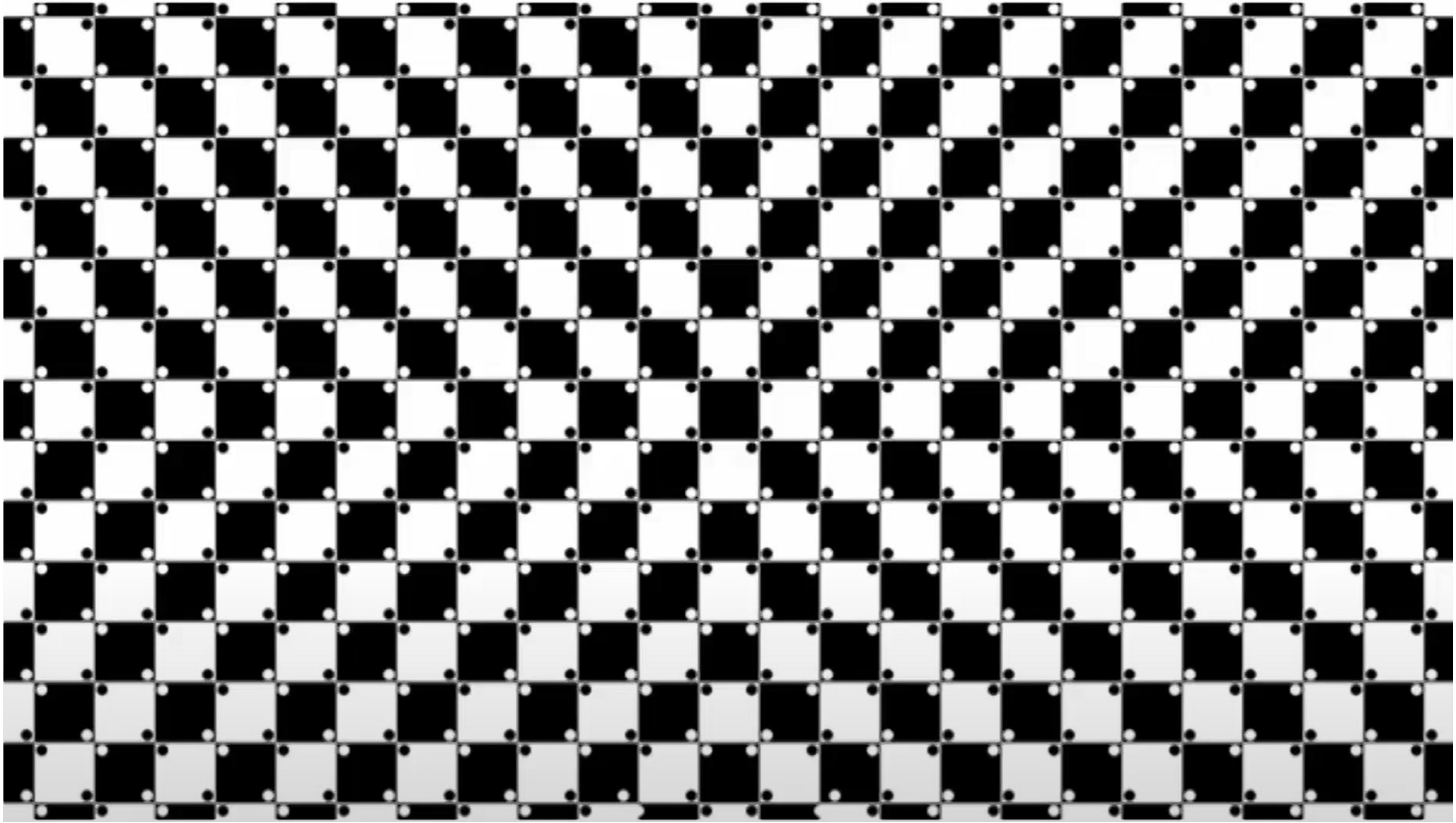

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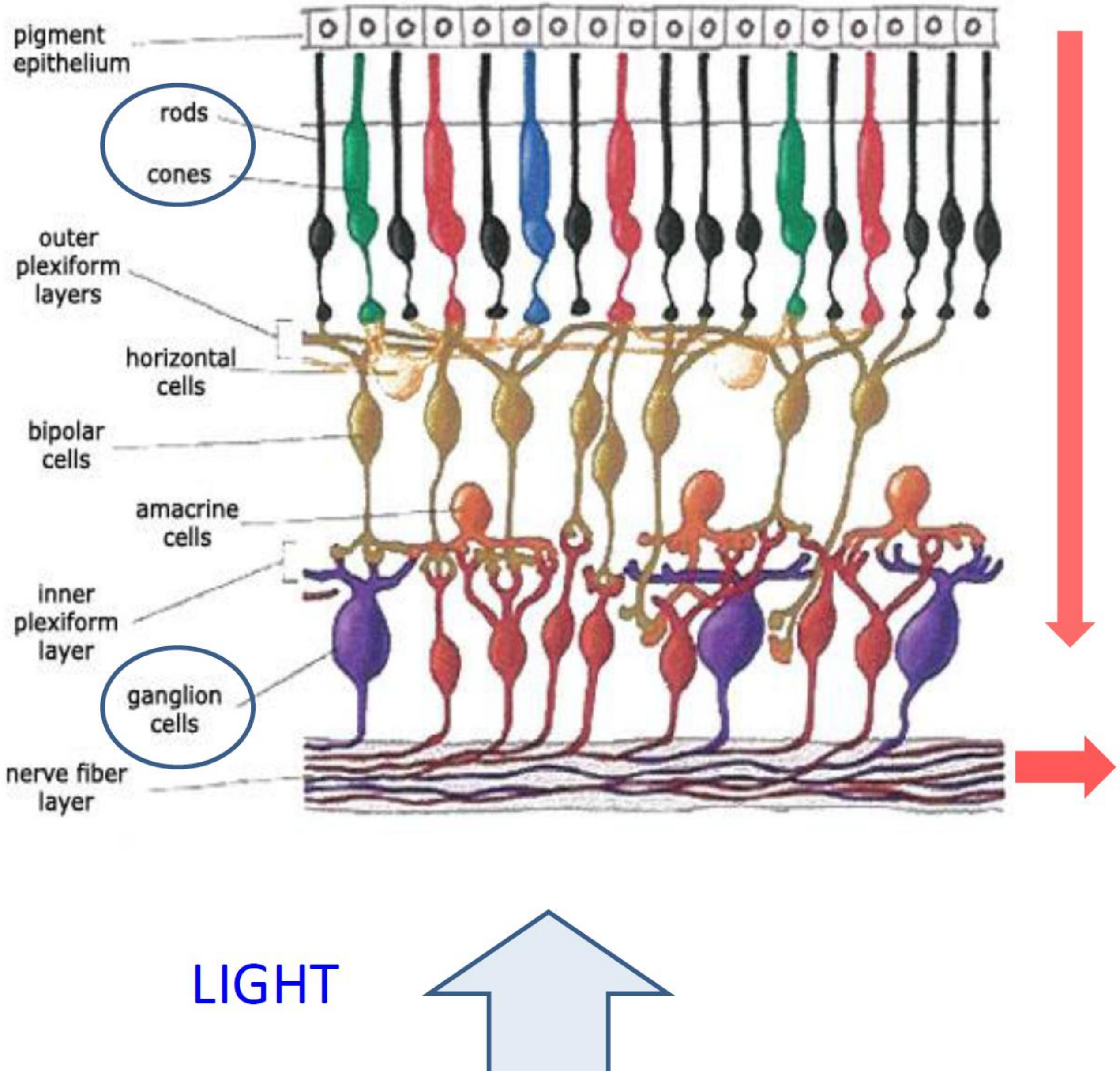
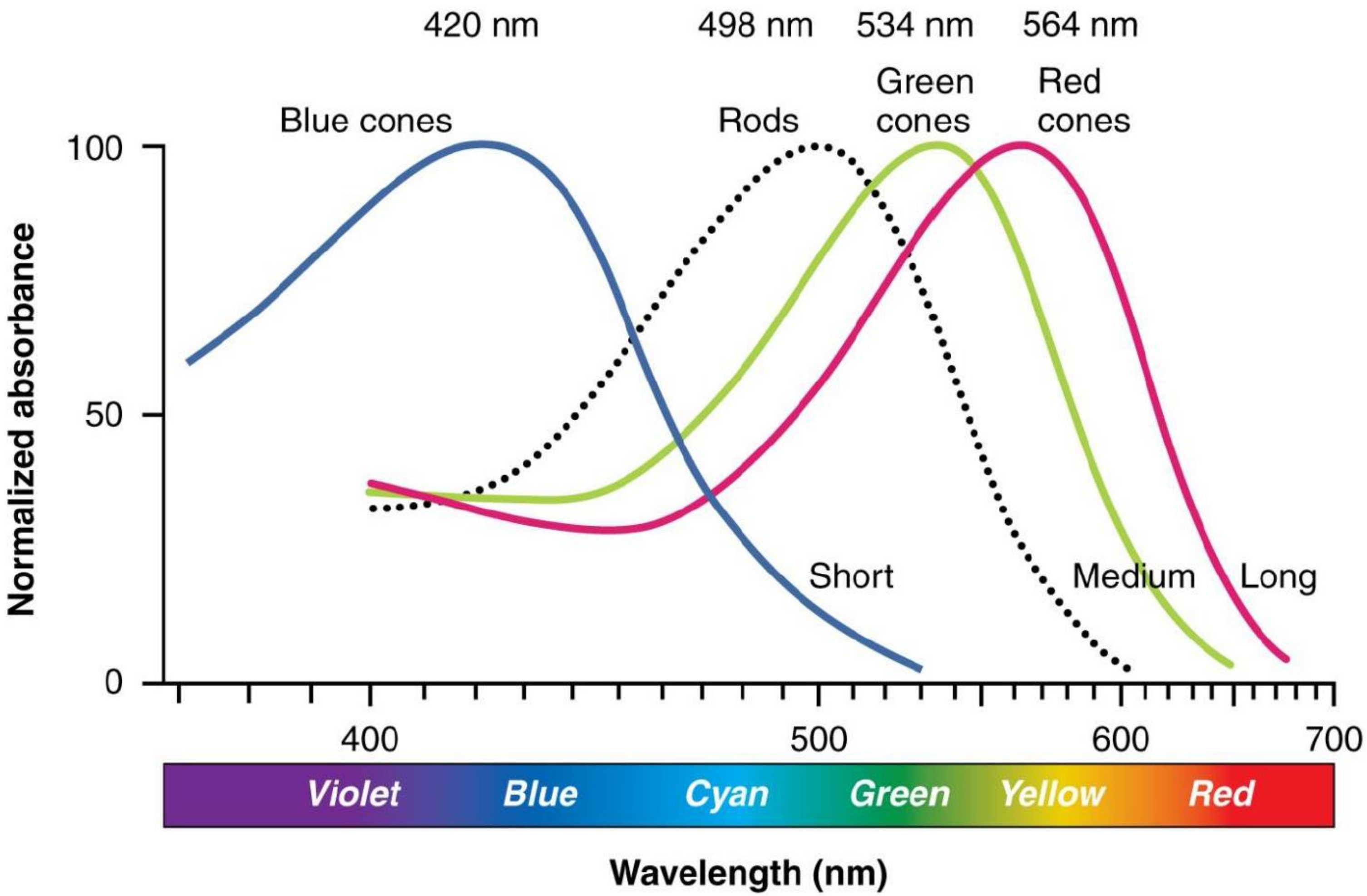
The retina



- The flow of electric signals goes from photoreceptors to ganglion cells.
- Axons of ganglion cells in the very back of the retina form the optical nerve, which travels to the Lateral Geniculate Nucleus (LGN) in the thalamus.

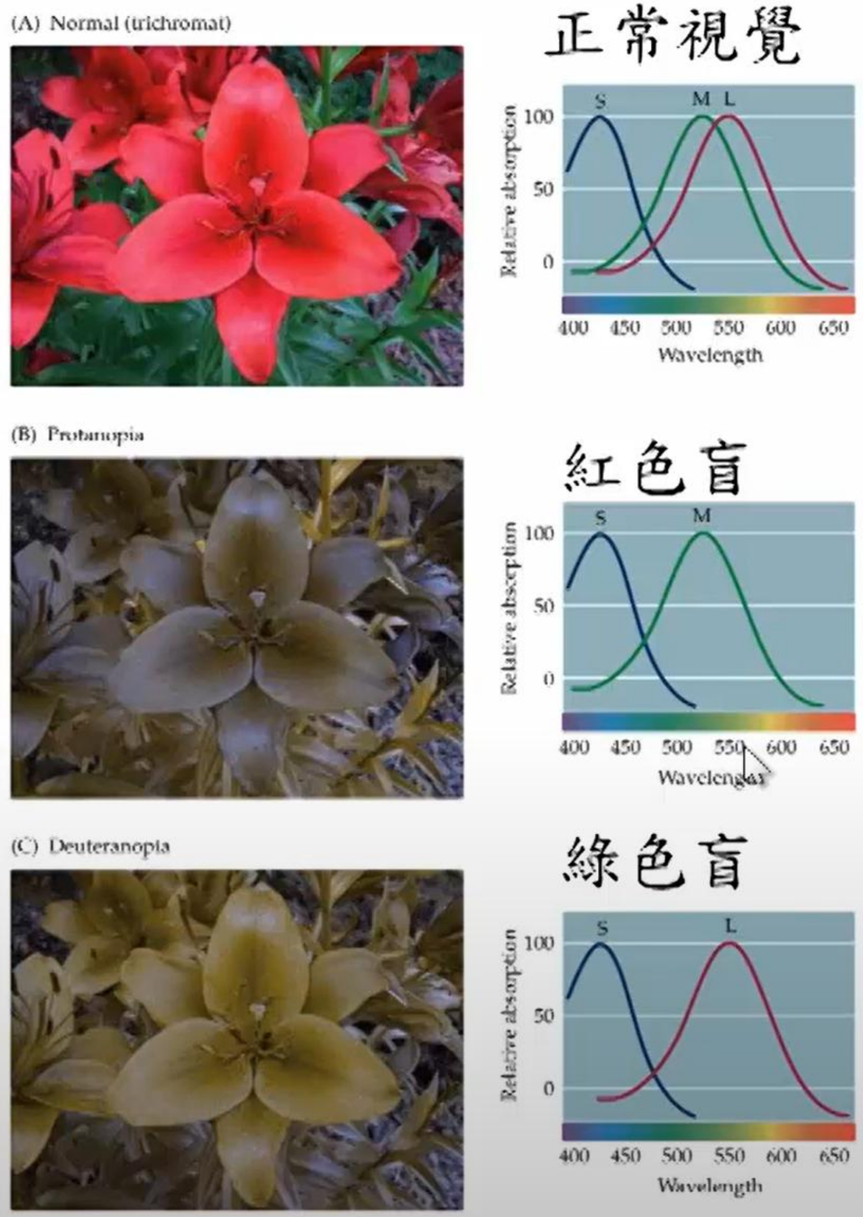
Light wavelength for cones and rods



- The spectrum of the wavelength for 3 types of cones and 1 type of rods.

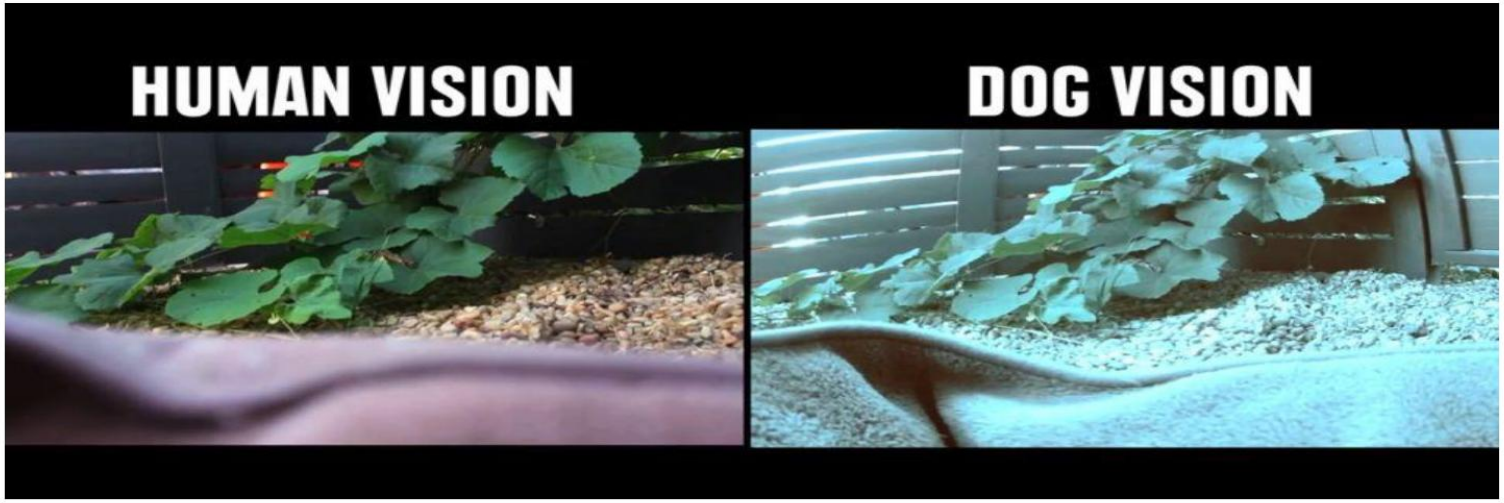
Deficits in visual perception

- Deficits in color perception:
  - ➢ Dichromats:
    - ✓ Red-green color-blind: medium/long wavelength missing
    - ✓ Blue-yellow color-blind: short wavelength missing
    - ✓ Normal cortical visual areas

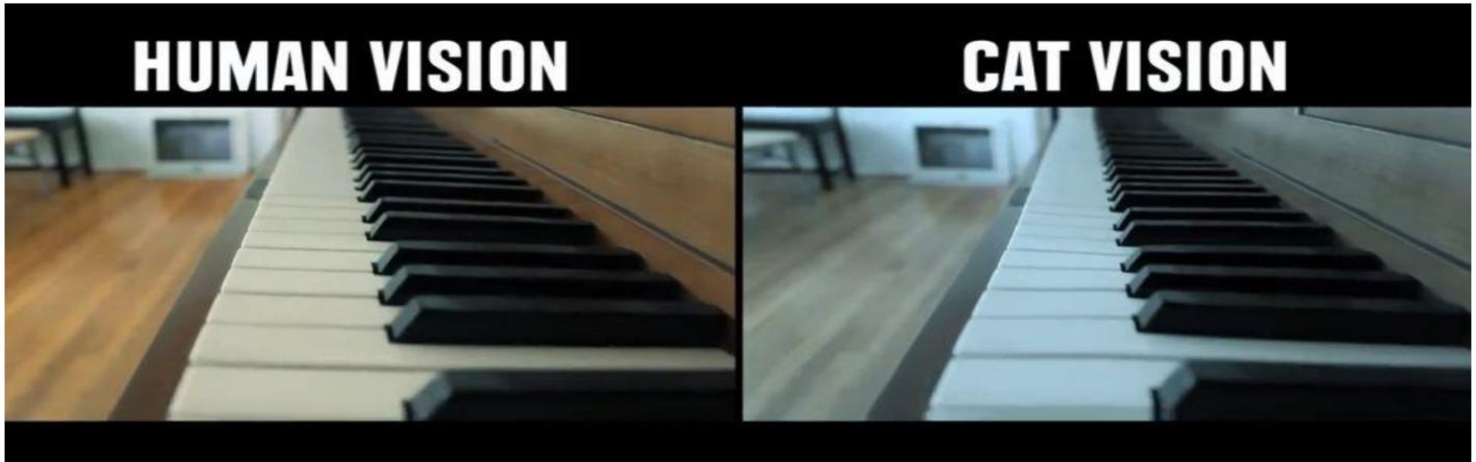


Photoreceptor sensitivity

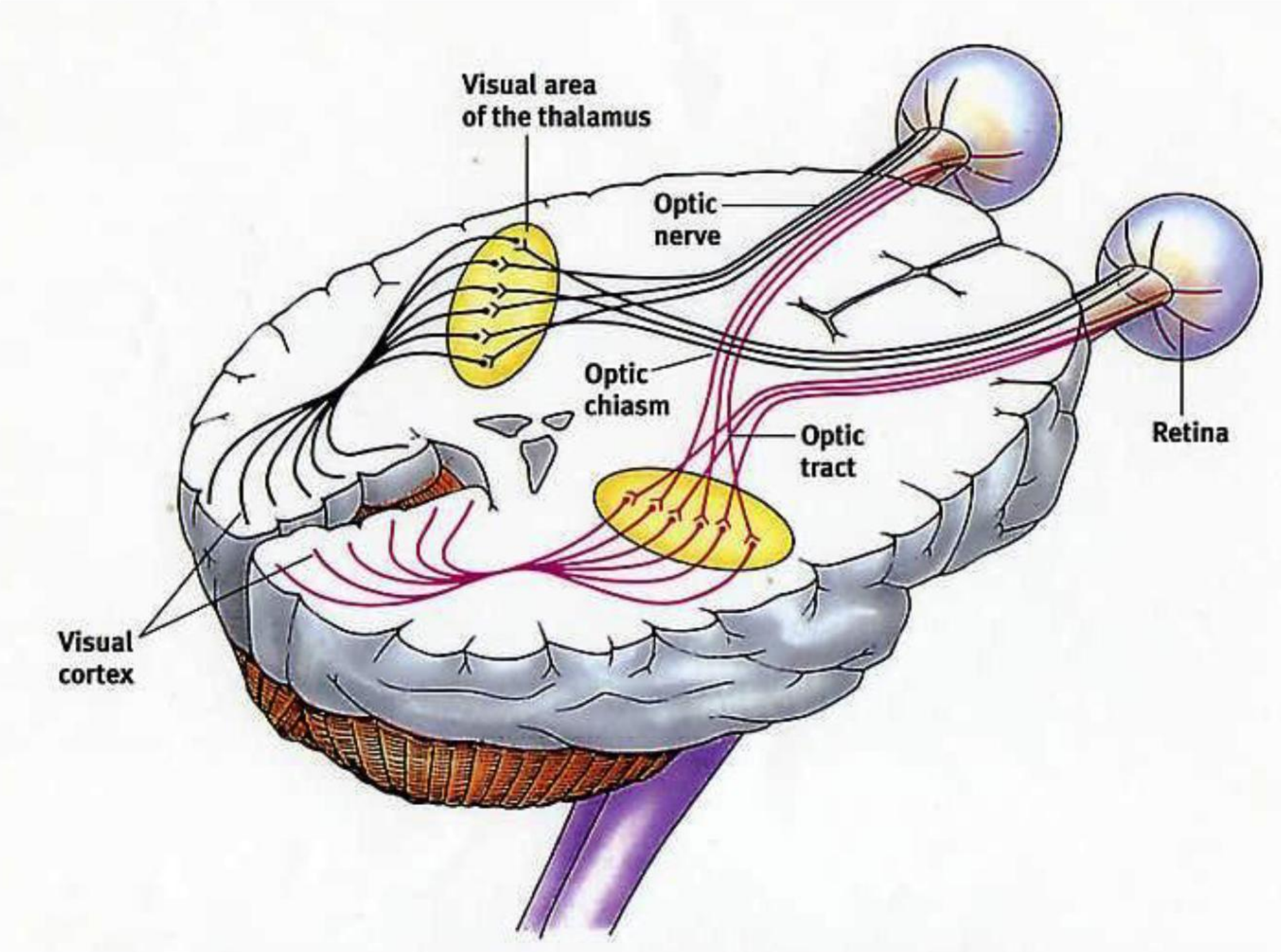
- Dogs have only 2 cones, blue and yellow. They see like colour blind people.



- Cats have many more rods, thus an excellent night vision, but have fewer cones than humans, thus less colourful vision .



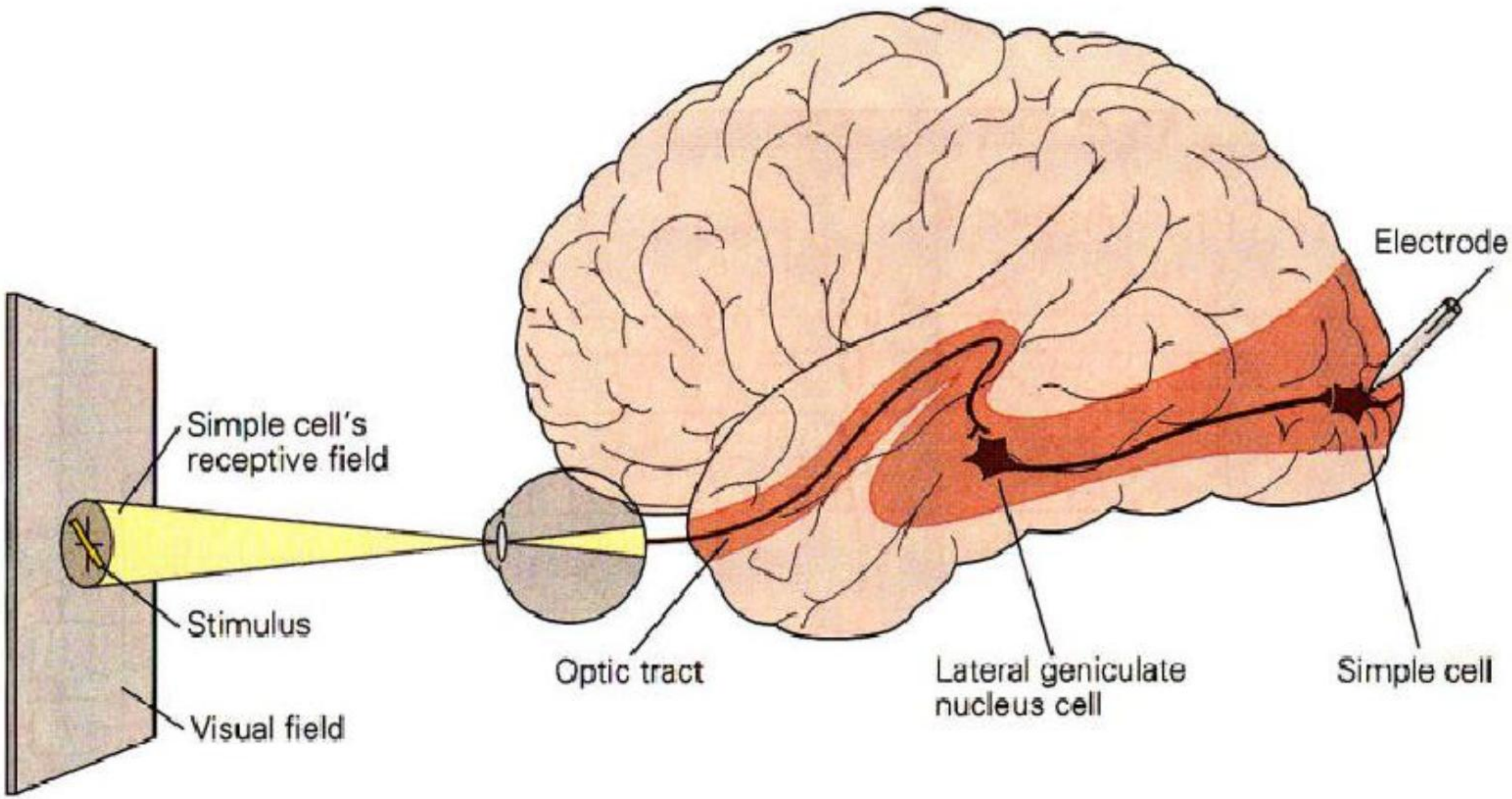
Optical tract in the human brain



- The signals in the form of trains of spikes go from the retinal ganglion cells to the LGN and then to the primary visual cortex V1.

What happens between the LGN and V1

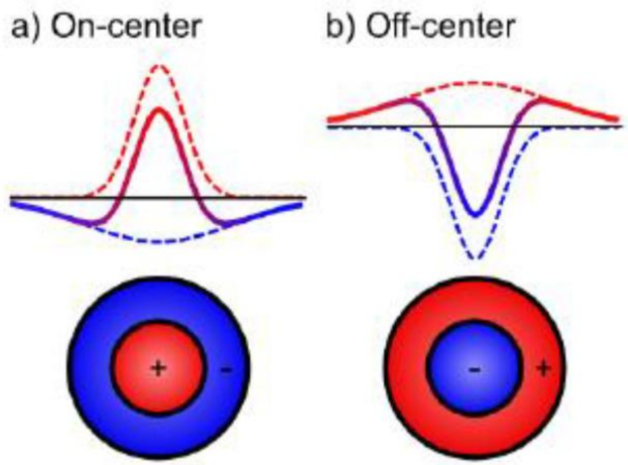
- Recording the neural activity in V1 in response to stimuli of different shapes revealed the V1 cells respond to the bar-like stimuli of light or **edges of different contrast** (Hubel and Wiesel).

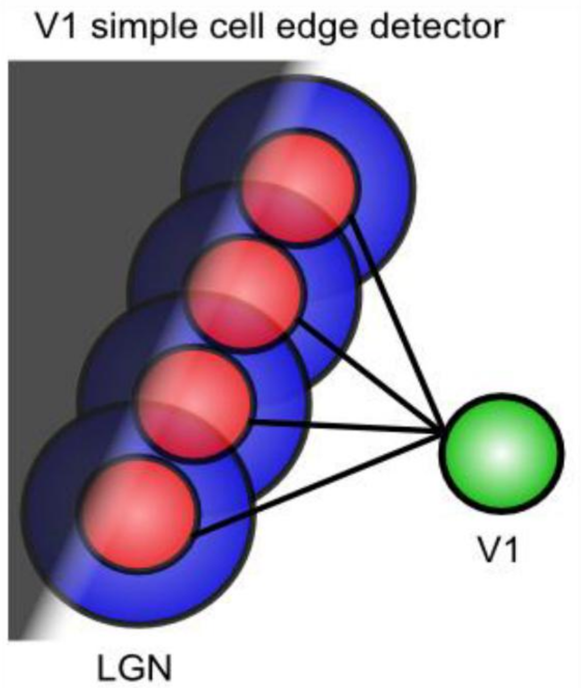


What happens between the LGN and V1

- The response properties of retinal cells can be summarized by the Difference-of-Gaussian (DoG) filters, with a narrow central region and a wider surround (also called centre-surround receptive fields).

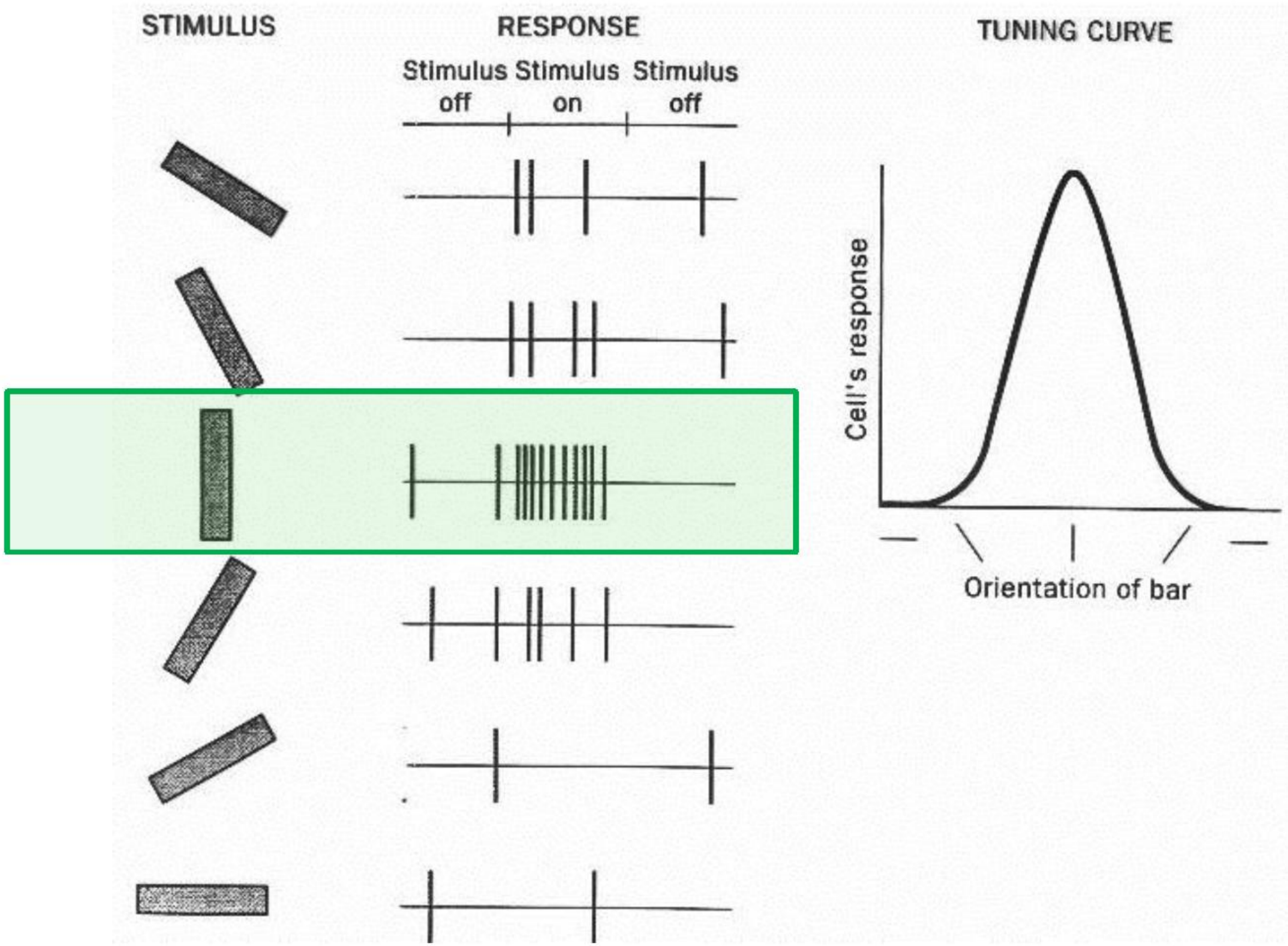
- A V1 simple cell detects an oriented edge of contrast in the image thanks to receiving spikes from a series of LGN on-centre cells aligned along the edge.





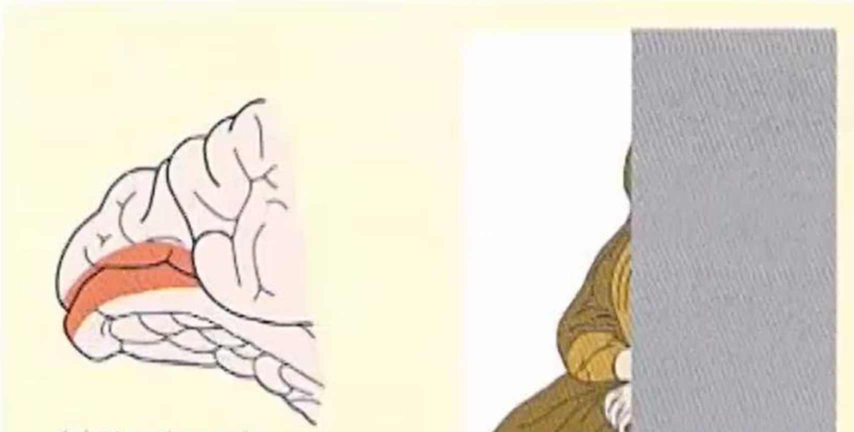
Orientation tuning data in V1 neurons

- Orientation tuning of an individual V1 neuron in response to bar-like stimuli at different orientations -- this neuron shows a preference for vertically oriented stimuli.



Deficits in visual perception

- Deficits in visual perception
  - Cortical blindness and "blindsight"
    - ✓ Lesions around V1
    - ✓ Hemianopia (half of visual field)
    - ✓ Quadrantanopia (1/4)
    - ✓ Scotoma (small region)



(a) Hemianopia

(c) Quadrantanopia

(b) Scotoma

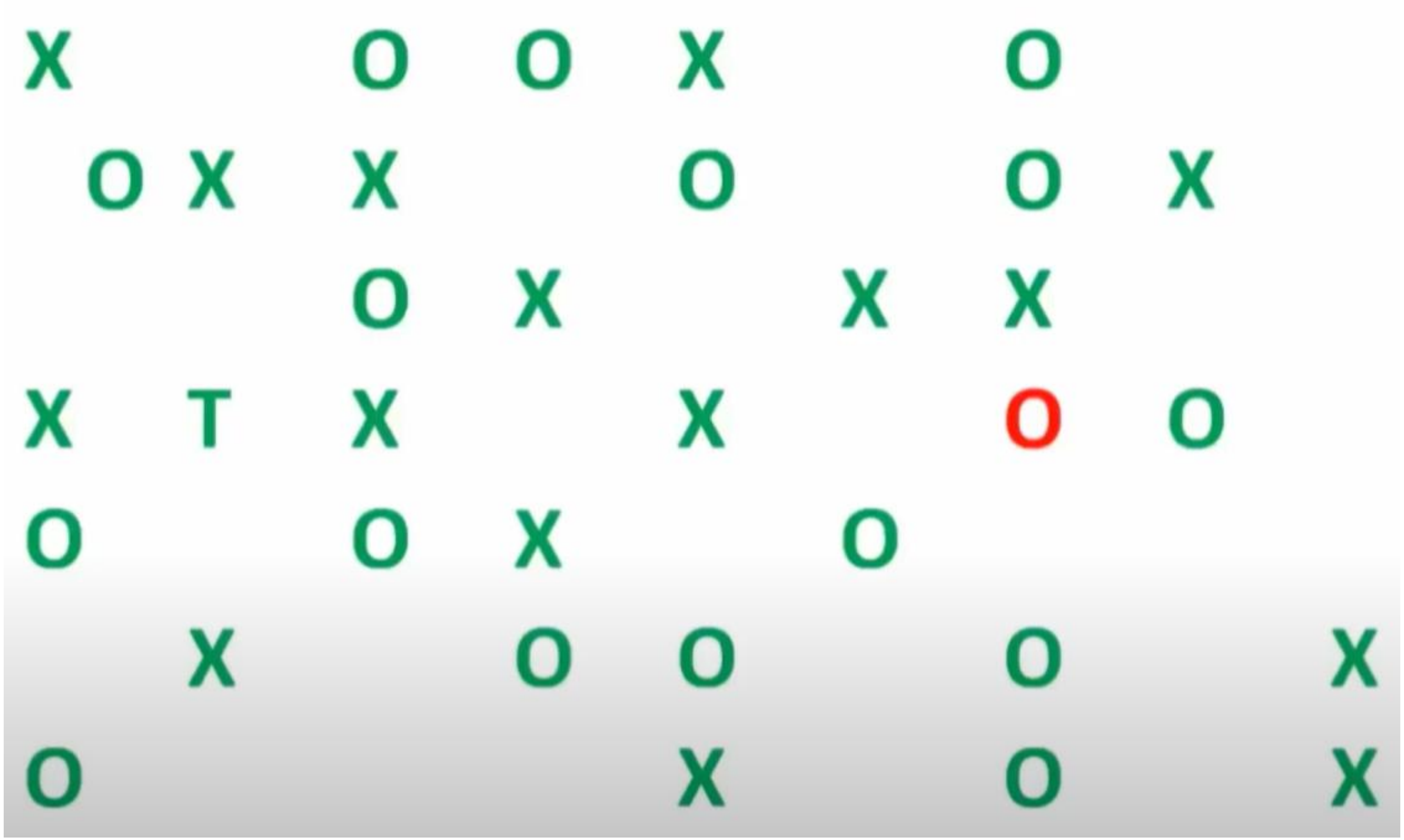
Deficits in visual perception

- Deficits in color perception
  - Achromatopsia (無色視覺)
    - ✓ "a": without, "chroma": hue
    - ✓ Lesions around V4 (color) area due to strokes and tumors

Deficits in visual perception

- Deficits in motion perception
  - ➢ Akinetopsia (motion blindness)
    - ✓ view the world as a series of snapshots
    - ✓ Deficits in MT area

- Pop-out search
  - target can be identified by a single feature



- Conjunction search
  - the target is defined by the conjunction of two or more stimulus features (e.g., the color red and the letter T)

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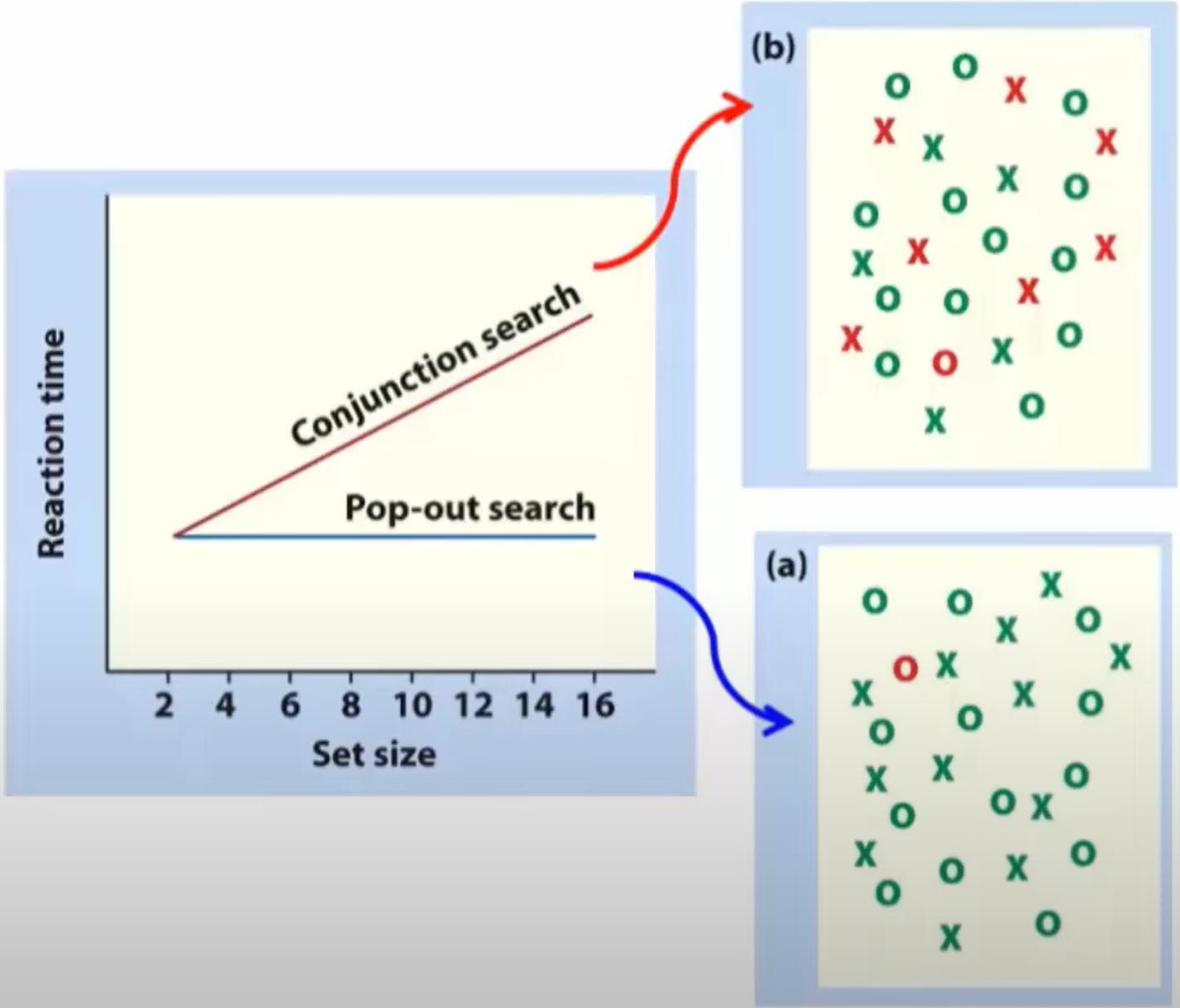
Pop-out search and conjunction search



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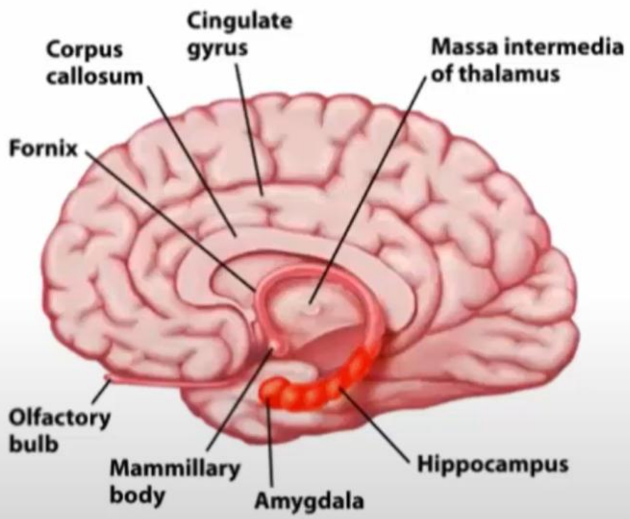
Different types of spatial representation

- Finding the nearest bank and picking up a pencil both require spatial processing
- Brain doesn't regard space as a continuous single entity
- Space in the brain exists in many forms:
  - ➢ Locations on sensory surfaces (e.g. retina; retinocentric space)
  - ➢ Location of objects relative to the body (egocentric space)
  - ➢ Location of objects relative to each other (allocentric space)

Different types of spatial representation

- Space
  - ➢ A common dimension of most perceptual systems and of motor/action system
  - ➢ e.g., a cat comes: by sight, sound, touch or smell.
- Cross-modal perception
  - ➢ Integrating information across sensory modalities



Space, attention and the parietal lobe

- Space and attention:
  - ➢ Attention tends to be directed to locations in space (space is a common dimension of different sensory systems and our motor system).
    - e.g., catch baseball
  - ➢ Attention may be needed to bind together different aspects of conscious perception (e.g. shape and color, sound and vision)
    - e.g., find faces and/or a vase

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A “spotlight” on attention

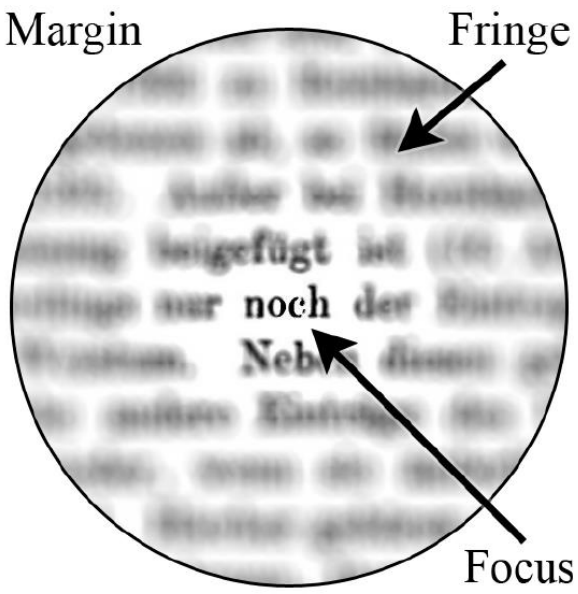
- The spotlight:
  - ➢ highlights a particular location in space (a salient object)
  - ➢ move from one location to another (e.g., eye movement)

Attention

- It is a cognitive process of selectively concentrating on a discrete aspect of information, whether subjective or objective, while ignoring other perceivable information.

- The term **"spotlight"** was introduced by William James, who described attention as having a focus, a margin, and a fringe. The focus is an area that extracts information from the visual scene with a high-resolution, the geometric center of which being where visual attention is directed.

- Surrounding the focus is the fringe of attention, which extracts information in a much more crude fashion (i.e., low-resolution).



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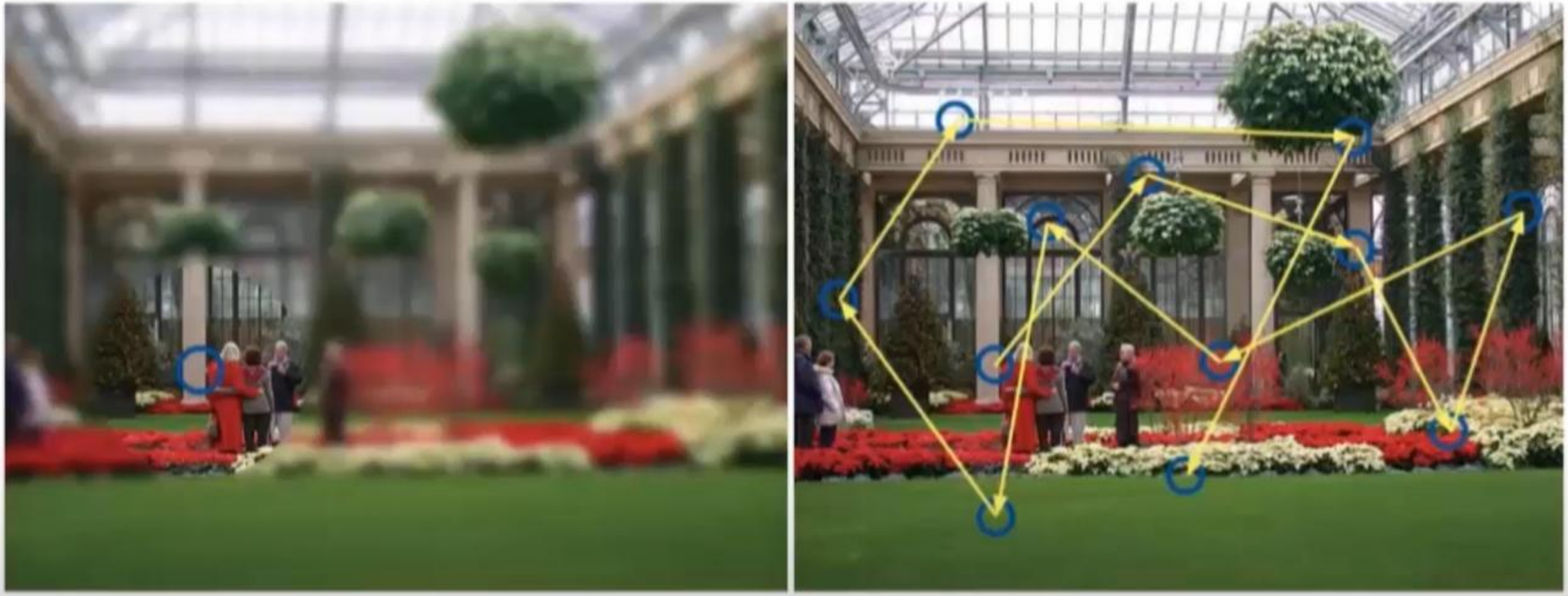
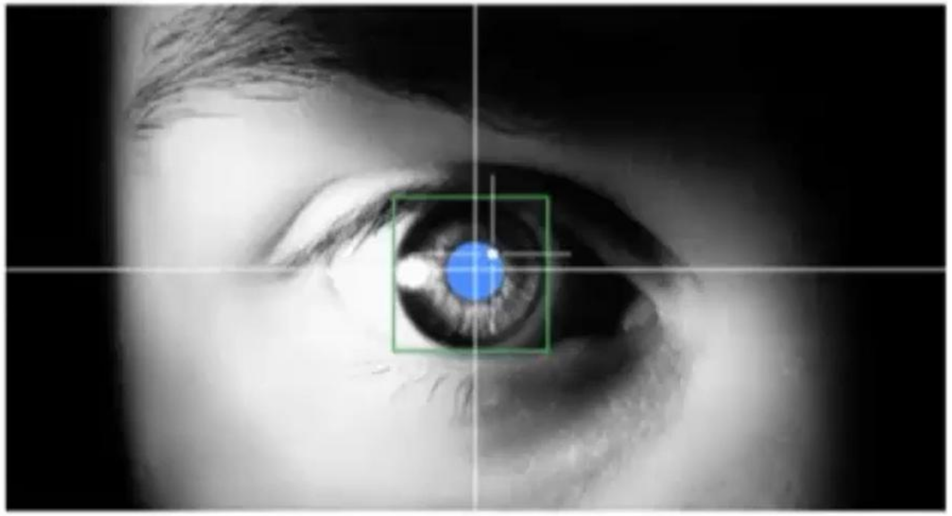
Eye movement:
fixation location and saccade



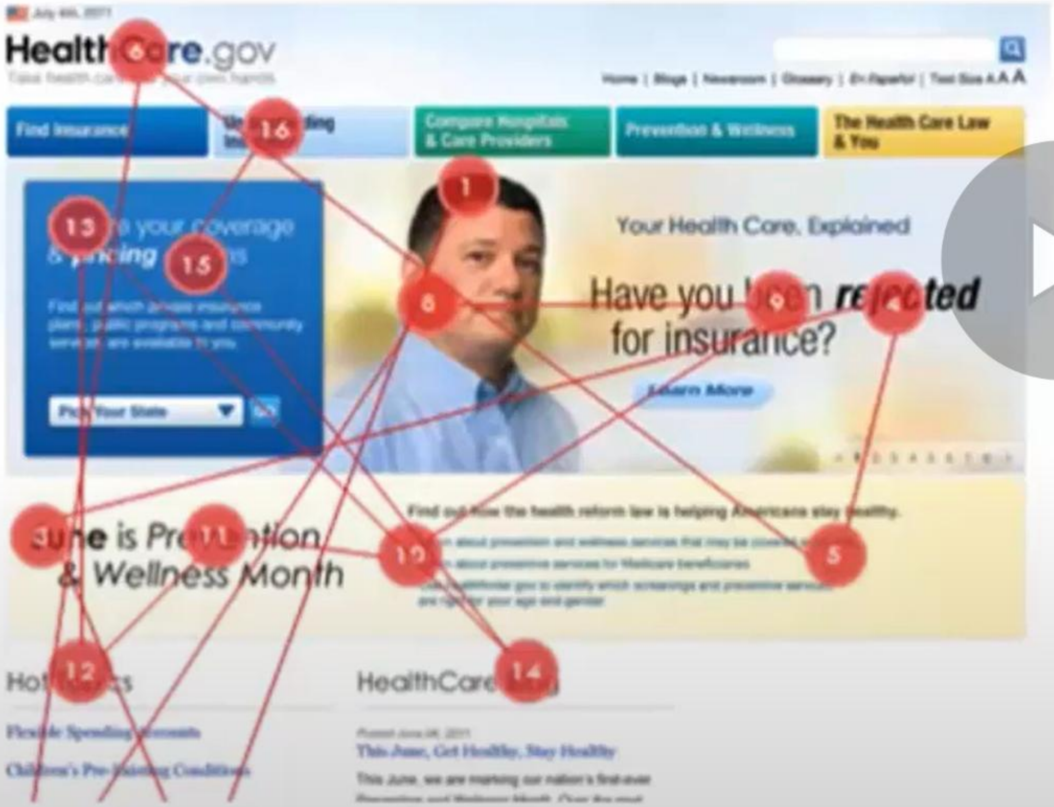
Eye-tracker

- ✓ Detecting pupil and corneal reflection
- ✓ Calibration: mapping eye movements to eye fixation positions

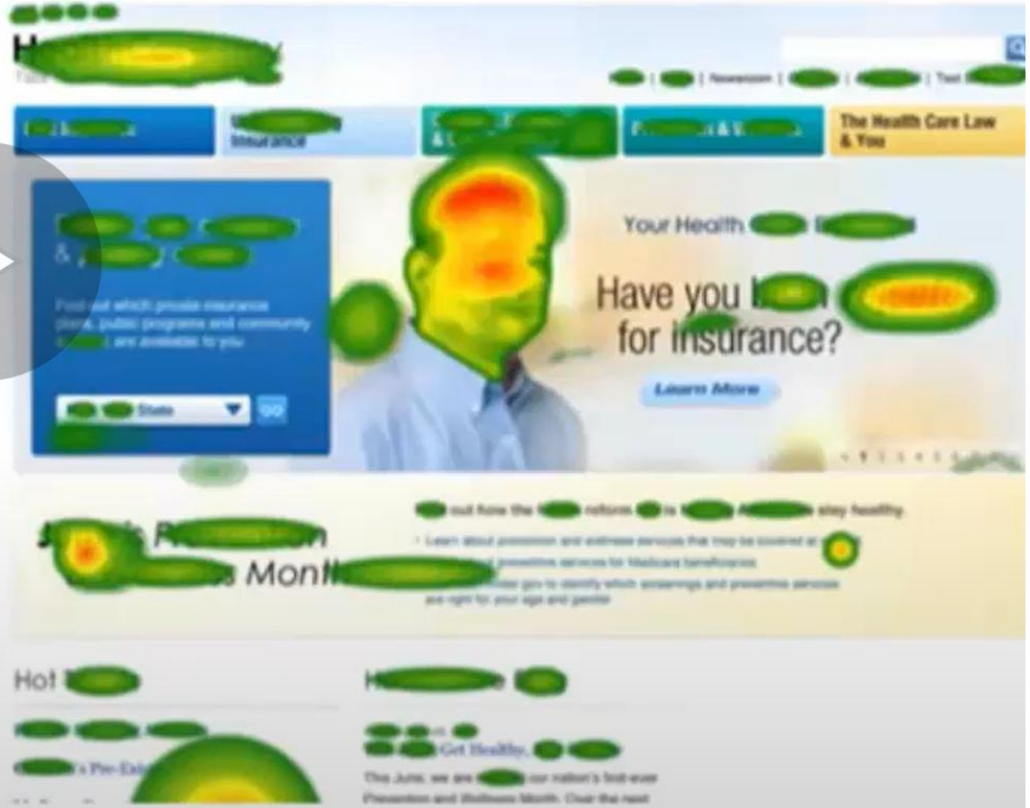


Eye-tracker

Eye movement map



Fixation map



http://is.gd/QZX5RF

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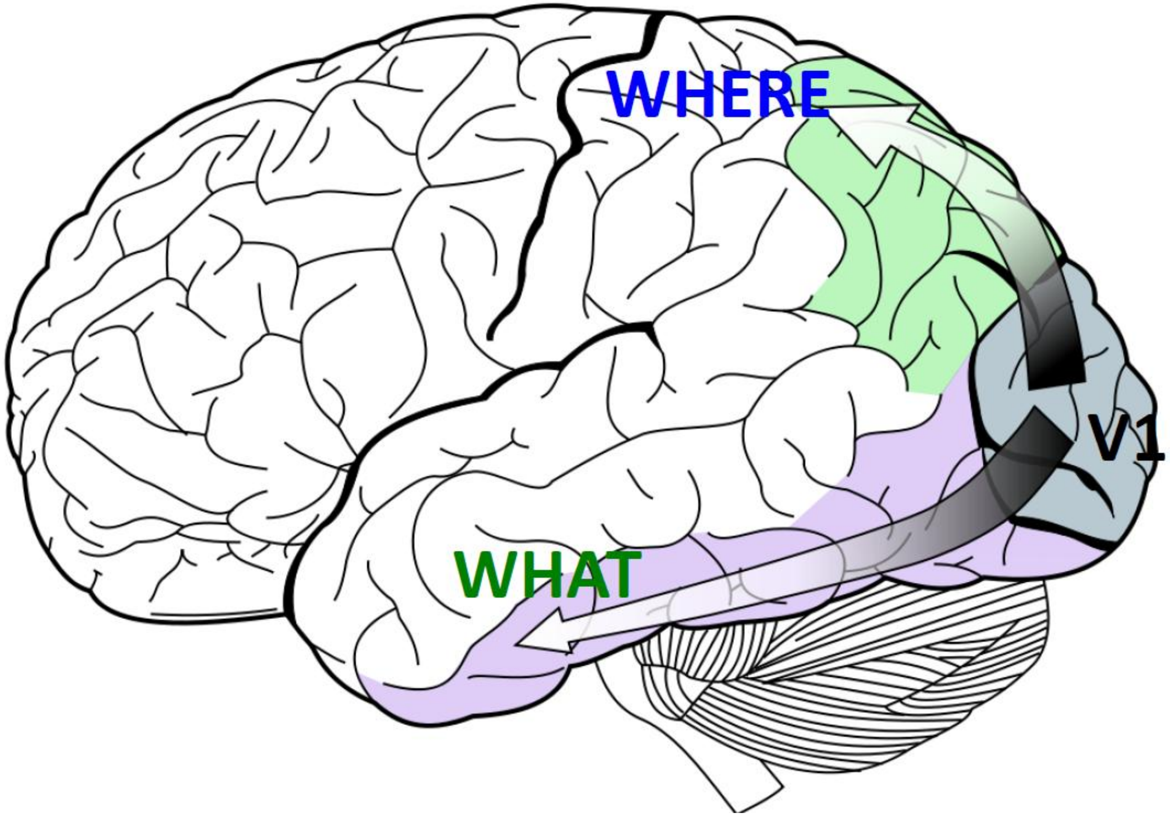
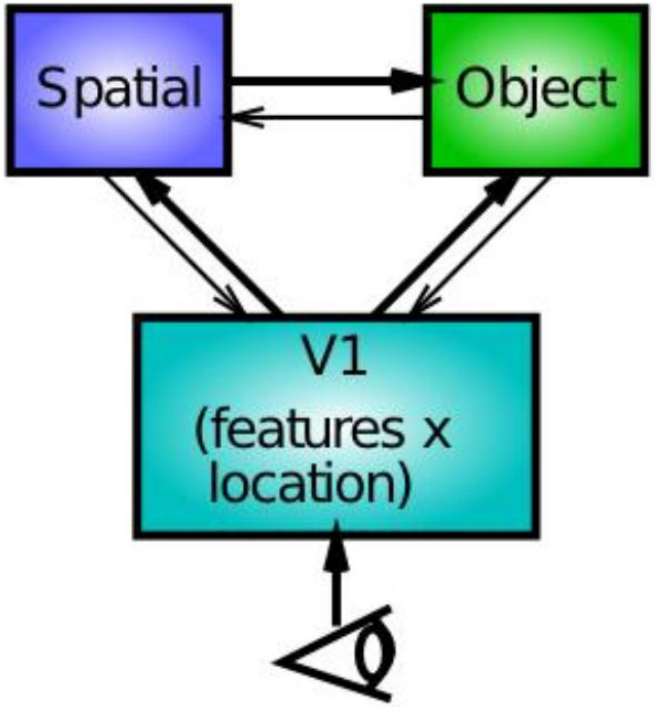
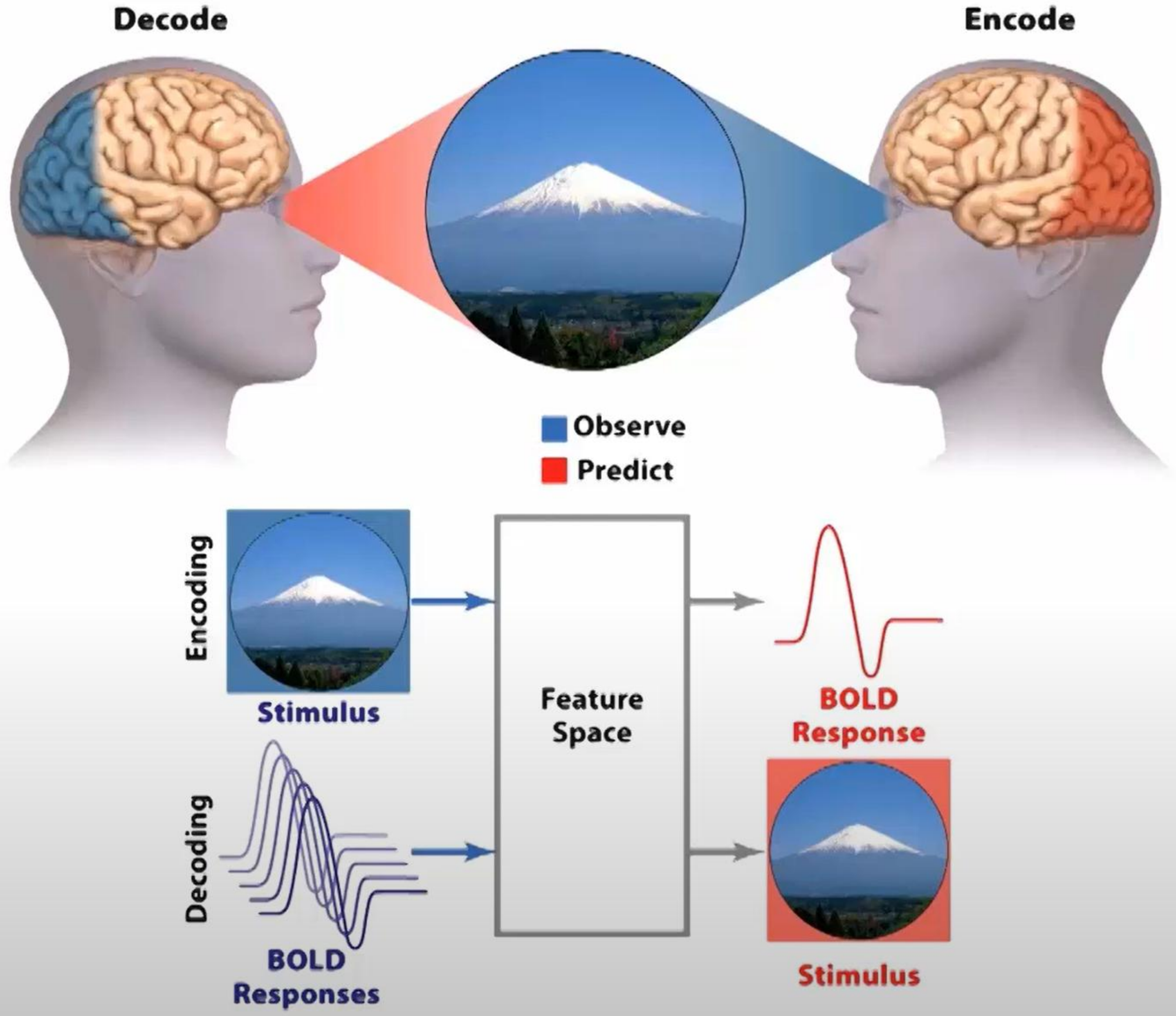
Summary

- The ventral stream (the "WHAT pathway") is involved with object visual identification and recognition. The dorsal stream (the "WHERE pathway") is involved with processing the object's spatial location relative to the viewer. They interact together.





Encoding vs. decoding



Different kinds of neural codes

- Rolls and Deco (2002) summarize three neural representations:
  - Local representation (= grandmother cells)
    - ✓ All the information about a stimulus/event is carried in one of the neurons.
  - Fully distributed representation
    - ✓ All the information about a stimulus/event is carried in all of the neurons of a given population.
  - Sparse distributed representation
    - ✓ A distributed representation in which a small proportion of the neurons carry information about a stimulus/event

Computational issues in motor controls

- Hierarchical representation of action sequences
  - ➢ High-level representation of the action is independent of any particular muscle group.
  - ➢ Motor representations are not linked to particular effector systems